# IES Electrical Engineering Topic wise Questions

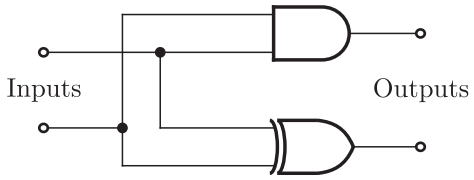
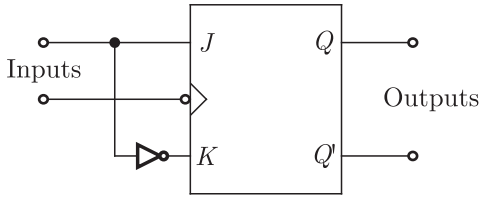
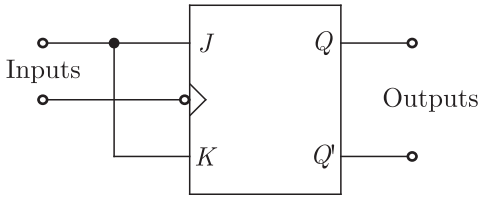
## Analog & Digital Circuits




**YEAR 2008**

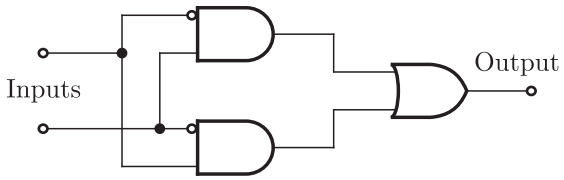
**MCQ 1**

Match **List I** (Logic circuit/function) with **List II** (Circuit realization) and select the correct answer using the code given below the lists :

| **List I** | | **List II** |
|---|---|---|
| A. D flip-flop | 1. |   |
| B. T flip-flop | 2. |   |
| C. Exclusive OR | 3. |   |

D. Half-adder

4.



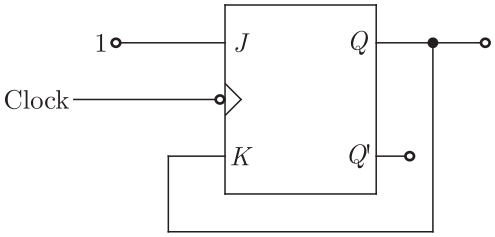

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 3 | 2 | 4 | 1 |
| (B) | 2 | 3 | 4 | 1 |
| (C) | 1 | 3 | 4 | 2 |
| (D) | 2 | 4 | 3 | 1 |

gate help

**MCQ 2**

In the circuit given in the below figure, $Q = 0$ initially. What shall be the subsequent states of $Q$ when clock pulses are given ?




(A) 1, 0, 1, 0,.....

(B) 0, 0, 0, 0,.....

(C) 1, 1, 1, 1,.....

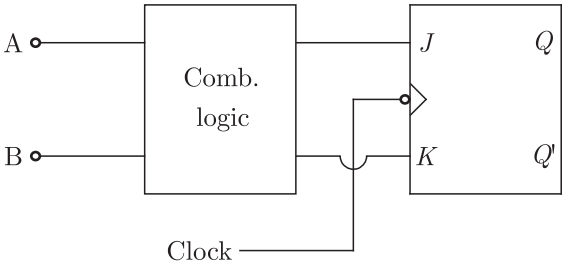
(D) 0, 1, 0, 1,.....

**MCQ 3**

The following truth table has to be realized with the circuit shown in the figure :

| A | B | $Q_{n+1}$ |
|---|---|---|

| | | |
|---|---|---|
| 0 | 0 | $Q'_n$ |
| 0 | 1 | 1 |
| 1 | 0 | $Q_n$ |
| 1 | 1 | 0 |




What is the output of the combinational logic circuit to the $J$ input ?

(A) $\overline{AB}$

(B) $\overline{A}$

(C) $\overline{B}$

(D) $AB$

gate help

**MCQ 4**

A J-K flip-flop can be made from an S-R flip-flop by using two additional

(A) NAND gates.

(B) OR gates.

(C) NOT gates.

(D) NOR gates.

**MCQ 5**

Match **List I** (Semiconductor technology) with **List II** (Characteristic) and select the correct answer using the code given below the lists :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | TTL | 1. | Maximum power consumption |
| B. | ECL | 2. | Highest packing density |
| C. | NMOS | 3. | Least power consumption |
| D. | CMOS | 4. | Saturated logic |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 1 | 4 | 2 | 3 |
| (B) | 4 | 1 | 2 | 3 |
| (C) | 1 | 4 | 3 | 2 |
| (D) | 4 | 1 | 3 | 2 |

**MCQ 6**

The maximum junction-temperature of a transistor is 150°C and the ambient temperature is 25°C. If the total thermal impedance is 1°C/W, what is the maximum power dissipation ?

(A) 1/175 W

(B) 175 W

(C) 125 W

(D) 1/125 W

**MCQ 7**

For the circuit shown in figure A the $V$-$I$ (voltage-current) characteristic of the circuit using ideal components is given by curve a in figure B.

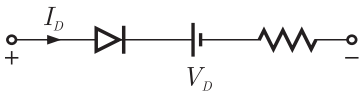



figure A

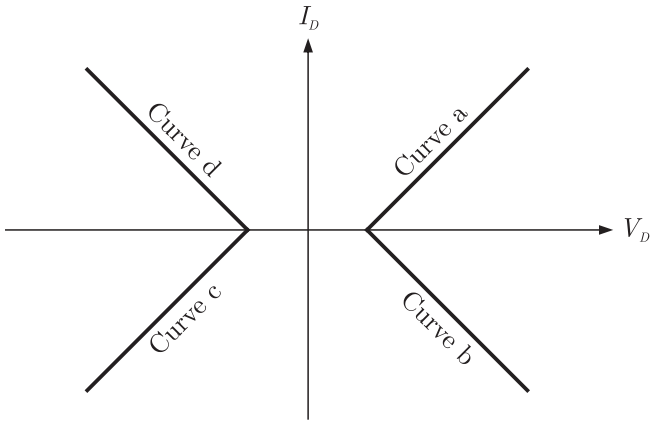



figure B

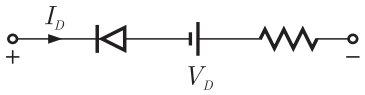



figure C

Which curve in figure B represents the $V$-$I$ characteristics for the circuit shown in figure C ?

(A) Curve $a$

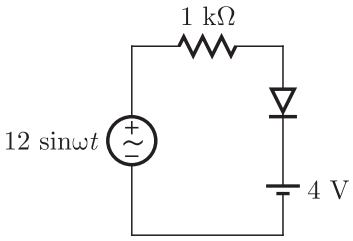
(B) Curve $b$

(C) Curve $c$

(D) Curve $d$

**MCQ 8**

What is the peak current through the resistor in the circuit given below assuming the diode to be ideal ?




(A) 4 mA

(B) 8 mA

(C) 12 mA

(D) 16 mA

**MCQ 9**

For a rectifier circuit, percentage voltage regulation is equal to which one of the following ?

(A) $\frac{V_{\text{no load}} - V_{\text{full load}}}{V_{\text{no load}}} \times 100$

(B) $\frac{V_{\text{no load}} - V_{\text{full load}}}{V_{\text{full laod}}} \times 100$

(C) $\frac{V_{\text{no load}} - V_{\text{full load}}}{V_{\text{no load}} + V_{\text{full load}}} \times 100$

(D) $\frac{V_{\text{full load}}}{V_{\text{no load}}} \times 100$

**MCQ 10**

**Assertion (A) :** In television transmission, interlaced scanning is used.

**Reason (R) :** Interlaced scanning provides increased picture brightness.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is not the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

**MCQ 11**

Consider the following statements in Johnson counter :

1. A MOD-6 Johnson counter requires 3 FFs.
2. Johnson counter requires decoding gates.
3. To decode each count, one logic gate is used. Each gate requires only two inputs regardless of the number of FFs.

Which of the statements given above are correct :

(A) 1 and 2

(B) 2 and 3

(C) 1 and 3

(D) 1, 2 and 3

**MCQ 12**

What is the simplified form of the Boolean expression

$$T = (X + Y)(X + \overline{Y})(\overline{X} + Y)$$

(A) $\overline{X}\,\overline{Y}$

(B) $\overline{X}Y$

(C) $XY$

(D) $X\overline{Y}$

**MCQ 13**

Match **List I** (Expression-I) with **List II** (Expression-II) and select the correct answer using the code given below the lists :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | $ABC + AB\overline{C} + A\overline{B}C$ | 1. | $\overline{A} + BC$ |

| | | | |
|---|---|---|---|
| B. | $\overline{A}B\overline{C}+AB\overline{C}+B\overline{C}$ | 2. | $A(B+C)$ |
| C. | $\overline{A}BC+A\overline{B}C$ | 3. | $B\overline{C}+AB\overline{C}+ABC$ |
| D. | $\overline{A\overline{B}}+\overline{A}B+ABC$ | 4. | $AB+BC+AC$ |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 2 | 1 | 4 | 3 |
| (B) | 4 | 3 | 2 | 1 |
| (C) | 2 | 3 | 4 | 1 |
| (D) | 4 | 1 | 2 | 3 |

### MCQ 14

The AND function can be realized by using only $n$ number of NOR gates. What is $n$ equal to ?

(A) 2

(B) 3

(C) 4

(D) 5




### MCQ 15

The Boolean expression $A.B+\overline{A}.\overline{B}$ is logically equivalent to which of the following ?

1. $(A+\overline{B}).(\overline{A}+B)$
2. $(\overline{A}+\overline{B}).(A+B)$
3. $\overline{(A.\overline{B}+\overline{A}.B)}$
4. $\overline{(A.B)}.\overline{(\overline{A}.\overline{B})}$

Select the correct answer using the code given below :

(A) 1 and 2

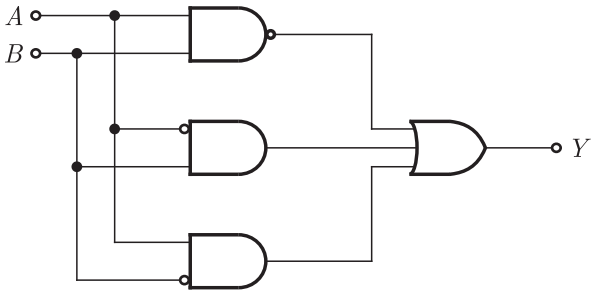
(B) 2 and 3

(C) 1 and 3

(D) None of the above

### MCQ 16

In the given circuit, the output $Y$ equals which one of the following ?




(A) $A + B$

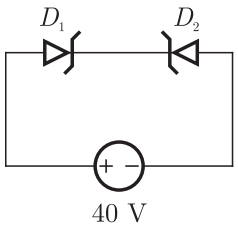
(B) $\overline{A}B + A\overline{B}$

(C) AB

(D) $\overline{A} + \overline{B}$

**MCQ 17**

In the circuit given below the Zener diode $D_1$ has a reverse breakdown voltage of 100 V and reverse saturation current of $25\,\mu A$. The corresponding values for $D_2$ are 50 V and $50\,\mu A$. What is the current in the circuit ?




(A) $25\,\mu A$ anticlockwise

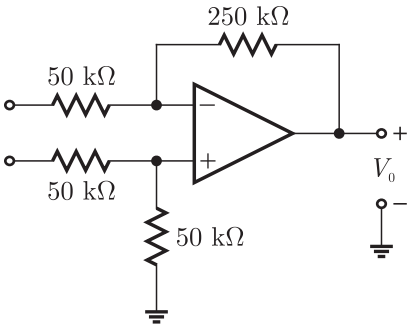
(B) $25\,\mu A$ clockwise

(C) $50\,\mu A$ anticlockwise

(D) $50\,\mu A$ clockwise

**MCQ 18**

What is the output voltage $V_0$ of the given circuit ?




(A) $-5\,V_a + 2.5\,V_b$

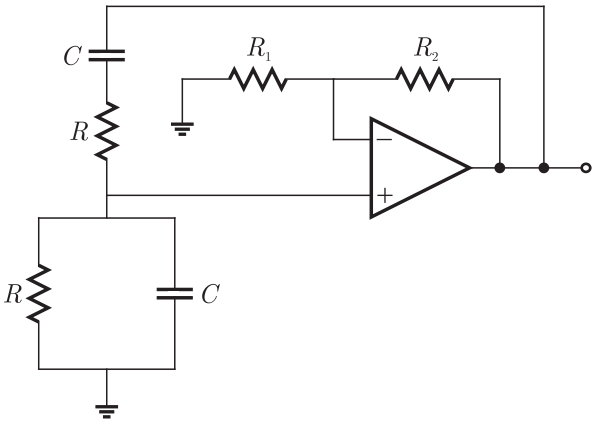
(B) $-5\,V_a + 3\,V_b$

(C) $-2.5\,V_a + 2.5\,V_b$

(D) $-2.5\,V_a + 3\,V_b$

**MCQ 19**

Consider the following statements in respect of the Wien bridge oscillator shown in the figure below :




1. For $R = 1$ kiloohm
   $C = \left(\frac{1}{2\pi}\right)\mu\text{F}, f = 1$ kHz
2. For $R = 3$ kiloohms
   $C = \left(\frac{1}{18\pi}\right)\mu\text{F}, f = 3$ kHz

Which of the statements given above is/are correct ?

(A) 1 only

(B) 2 only

(C) Both 1 and 2

(D) Neither 1 nor 2

**MCQ 20**

Consider the following statements :

1. When bridge oscillator is suitable for generating 1 kHz.
2. Colpitts oscillator is suitable for generating 1 MHz.

Which of the statements given above is/are correct ?

(A) 1 only

(B) 2 only

(C) Both 1 and 2

(D) Neither 1 nor 2




**MCQ 21**

A sinusoidal signal of 100 Hz is applied to an amplifier. The output current is

$$i_0 = 20\sin(628t) + 2\sin(1256t) + 1\sin(2512t)$$

What is the approximate percentage increase in power due to distortion ?

(A) 1.15

(B) 1.25

(C) 1.30

(D) 1.50

**MCQ 22**

A resistance $R_f$ is connected across the collector an : base of a BJT amplifier of gain $-A\,(A > 0)$. The input impedance of the amplifier will consist of transistor internal resistance $r_{b'e}$ shunted by which one of the following ?

(A) $R_f(1 + A)$

(B) $R_f(1 - A)$

(C) $R_f/(1 + A)$

(D) $R_f/(1 - A)$

**MCQ 23**

A negative feedback amplifier with open-loop gain $\frac{-A_0}{1+j\frac{\omega}{\omega_0}}$

$A_0 > 0$ and feedback factor $\beta(>0)$ will have a 3 dB cut-off at what frequency ?

(A) $\omega_0 A_0 \beta$

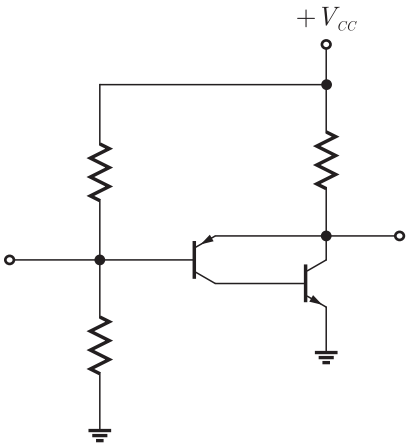
(B) $\omega_0(1 + A_0\beta)$

(C) $\omega_0/(1 + A_0\beta)$

(D) $\omega_0(1 - A_0\beta)$

**MCQ 24**

What is the transistor combination shown in the figure given below ?




(A) A Darlington pair.

(B) A complementary pair.

(C) It effectively acts as a single $p$-$n$-$p$ transistor.

(D) It effectively acts as a single $n$-$p$-$n$ transistor.

**MCQ 25**

What is the effect of cascading the amplifier stages ?

(A) To increase the voltage gain and increase the bandwidth.

(B) To increase the voltage gain and reduce the bandwidth.

(C) To decrease the voltage gain and increase the bandwidth.

(D) To decrease the voltage gain and reduce the bandwidth.

**MCQ 26**

**Assertion (A) :** It is not possible to design a current source using operational amplifier.

**Reason (R) :** Operational amplifier is a voltage-controlled voltage source.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is not the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

## YEAR 2007

**MCQ 27**

What is the value of LSB of an 8 bit DAC for 0-12.8 V range ?

(A) 1.6 V

(B) 50 mV

(C) 0.625 V

(D) 1.28 V

**MCQ 28**

A transistorised transformer coupled class-A amplifier supplied 0.94 W to a 4 kΩ load. The zero single dc-collector current is 31 mA, and the dc collector current with signal is 34 mA. What is the percent-second-harmonic distortion ?

(A) ~ 50%

(B) 0%

(C) ~ 20%

(D) Cannot be computed with the available information.

**MCQ 29**

Which of the following characteristics are possessed by a transformer-coupled class B push-pull power amplifier ?

1. It eliminates even harmonic distortion.
2. It suffers from cross-over distortion.
3. Its device ratings are higher than those of class A power amplifier.

4. Its collector circuit efficiency is more than that of class C power amplifier.

Select the correct answer using the code given below :

(A) 1, 3 and 4

(B) 2 and 4

(C) 1 and 2

(D) 3 and 4

**MCQ 30**

Which of the following are true for $h$-parameters of transistors ?

1. They are real numbers at audio frequencies.
2. They are easy to measure.
3. They vary widely with temperature.

Select the correct answer using the code given below :

(A) 1 and 2 only.

(B) 2 and 3 only.

(C) 1 and 3 only.

(D) 1, 2 and 3.

gate help

**MCQ 31**

Match **List I** (Semiconductor Property) with **List II** (Corresponding Unit) and select the correct answer using the code given below the lists :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Carrier mobility | 1. | eV (electron volt) |
| B. | Diffusion length | 2. | $m^2$/V-sec |
| C. | Diffusion-coefficient | 3. | m |
| D. | Energy gap | 4. | $m^2$/s |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 4 | 2 | 3 | 1 |
| (B) | 2 | 3 | 4 | 1 |
| (C) | 2 | 3 | 1 | 4 |

(D) 4 2 1 3

**MCQ 32**

If $\alpha_F$ and $\alpha_R$ denote the forward and inverted mode current gains of a BJT, which one of the following is correct ?

(A) $\alpha_F = \alpha_R$

(B) $\alpha_F < \alpha_R$

(C) $\alpha_F \geq \alpha_R$

(D) $\alpha_F >> \alpha_R$

**MCQ 33**

In a source follower, consider the following statements :

1. The voltage gain of a source follower is always less than one.
2. It has some current gain and power gain.
3. Its output resistance can be made low.

Which of the statements given above are correct ?

(A) 1 and 2 only.

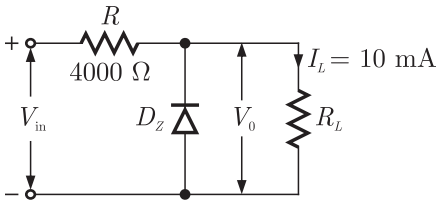
(B) 2 and 3 only.

(C) 1 and 3 only.

(D) 1, 2 and 3.

**MCQ 34**

A zener diode regulator shown in the figure given below is to be designed to meet the following specifications :




$I_L = 10\,\text{mA}, \text{V}_0 = 10\,\text{V}, \text{V}_{\text{in}}$ varier from 30 V to 50 V. The zener diode has $V_Z = 10\,\text{V}$ and $I_{zk}$ (knee current) = 1 mA. For satisfactory operation, which one of the following is correct ?

(A) $R \leq 1800\,\Omega$

(B) $2000\,\Omega \leq R \leq 2200\,\Omega$

(C) $3700\,\Omega \leq R \leq 4000\,\Omega$

(D) $R > 4000\,\Omega$

**MCQ 35**

For a half-wave rectifier, what is the output dc voltage ? (with peak voltage = $V_m$, dc current = $I_{DC}$ and forward resistance of diode = $R_f$)

(A) $V_{DC} = \frac{V_m}{\pi} + I_{DC} R_f$

(B) $V_{DC} = \frac{V_m}{\pi}$

(C) $V_{DC} = \frac{V_m}{\pi} - I_{DC} R_f$

(D) $V_{DC} = 0.707 \frac{V_m}{\pi} - I_{DC} R_f$

**MCQ 36**

Which of the following are coefficients for a regulated power supply (in the expression for change in output voltage) ?

1. Stability factor.
2. Output resistance.
3. Temperature coefficient.
4. Input resistance.

Select the correct answer using the code given below :

(A) 1 and 4 only.

(B) 1, 3 and 4.

(C) 2 and 3 only.

(D) 1, 2 and 3.

**MCQ 37**

Intermediate (I) layer of PIN-diode imparts which one of the following features to a $p$-$n$ junction diode ?

(A) High reverse blocking capability.

(B) High forward current rating.

(C) Inverting capability.

(D) Poor turn off performance.

**MCQ 38**

Which one of the following is the correct statement ?

In a two quadrant converter working in the 1st and 2nd quadrants

(A) load current and load voltage are always positive.

(B) load current is always negative.

(C) load current can be positive or negative.

(D) load current and load voltage are always negative.

**MCQ 39**

In a single phase full wave controlled bridge rectifier, minimum output voltage and maximum output voltage are obtained at which conduction angles ?

(A) $0^\circ, 180^\circ$ respectively.

(B) $180^\circ, 0^\circ$ respectively.

(C) $0^\circ, 0^\circ$ respectively.

(D) $180^\circ, 180^\circ$ respectively.

**MCQ 40**

**Assertion (A) :** MOSFETs perform very well in parallel operation.

**Reason (R) :** MOSFET has smaller turn-off time.

(A) Both A and R are true and R is the correct explanation of A.

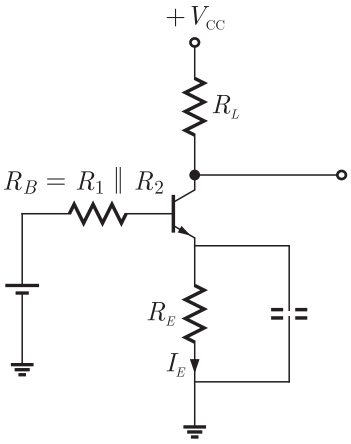
(B) Both A and R are true but R is not the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

**MCQ 41**

Fixed biasing of CE configuration is shown in the figure given below :




The current stabilization factor for $R_B << \beta$

$R_E$ is $S_i = 1 + \dfrac{R_B}{R_E}$

For $R_B << R_E$, what is the voltage stabilization factor $S_v$ ?

(A) $= \dfrac{R_E}{R_B + R_E}$

(B) $= \dfrac{R_B}{R_E}$

(C) $\approx -\dfrac{1}{R_E}$

(D) $\approx -\dfrac{R_E}{R_B + R_E}$

gate help

**MCQ 42**

Following transistor parameters were measured at $I_c = 5\,\text{mA}$, $V_{CE} = 10\,\text{V}$, $h_{fe} = 100$, $[A_1] = 10$ at 10 M rad/s. What is the value for $(C_e + C_c)$, if the unity-gain frequency $\omega_T = \dfrac{g_m}{C_e + C_c}$

(A) 20 nF

(B) 2 nF

(C) 5 pF

(D) 5 nF

**MCQ 43**

The function

$F = AB\overline{C} + ABC + \overline{A}BC + \overline{A}B\overline{C}$ ; can be reduced to which one of the following ?

(A) $F = A$

(B) $F = AB$

(C) $F = ABC$

(D) $F = B$

**MCQ 44**

An amplifier without feedback has a gain of 1000. What is the gain with a negative feedback of 0.009 ?

(A) 900

(B) 125

(C) 100

(D) 10




**MCQ 45**

Negative feedback in an amplifier leads to which one of the following ?

(A) Decrease in bandwidth.

(B) Increase in current gain.

(C) Increase in voltage gain.

(D) Decrease in voltage gain.

**MCQ 46**

Which of the following are the main advantages of class B push-pull power amplifier (using BJTs) ?

1. Even harmonics tend to cancel out at the output.
2. More power output per transistor.
3. Conversion efficiency can be as high as 78%.
4. Absence of cross-over distortion.

Select the correct answer using the code given below :

(A) 1, 2 and 3

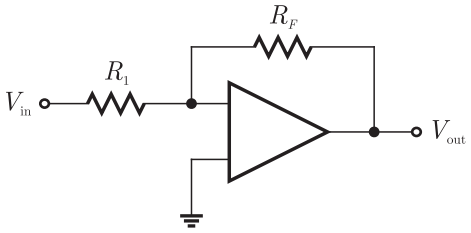
(B) 1, 2 and 4

(C) 1, 3 and 4

(D) 2, 3 and 4

**MCQ 47**

For the operational amplifier circuit shown in the figure below, what is the maximum possible value of $R_1$, if the voltage gain required is between $-10$ and $-25$ ? (The upper limit on $R_F$ is 1 MΩ)




(A) Infinity.

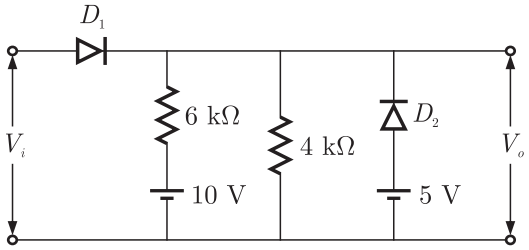
(B) 1 Mega ohm.

(C) 100 k ohm.

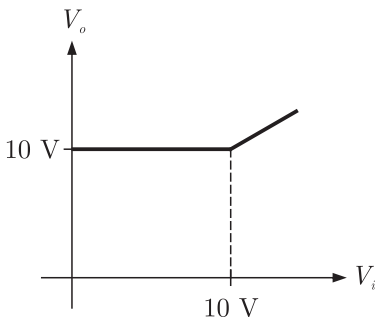
(D) 40 k ohm.

**MCQ 48**

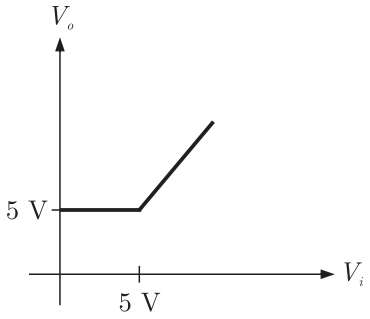
In the circuit given below, $D_1$ and $D_2$ are ideal. Which one of the following represents the transfer characteristics of the circuit ?



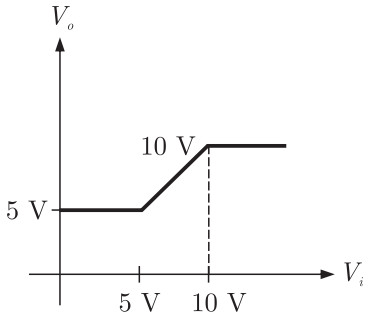

(A)



(B)

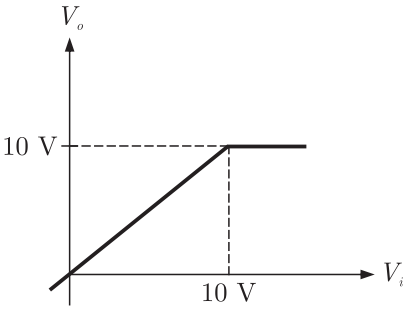



(C)



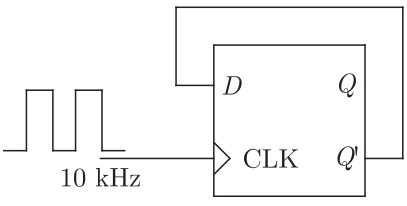

gate help

(D)




**MCQ 49**

The frequency of the clock signal applied to the rising edge triggered $D$ flip-flop shown in the below figure is 10 kHz. What is the frequency of the signal available at $Q$ ?




(A) 2.5 kHz

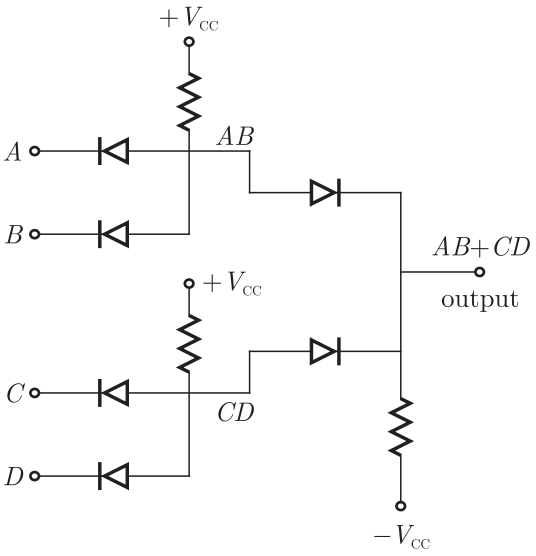
(B) 5 kHz

(C) 10 kHz

(D) 20 kHz

**MCQ 50**

What is the output of the gate circuit shown in the below figure ?




(A) $(A+B)(C+D)$

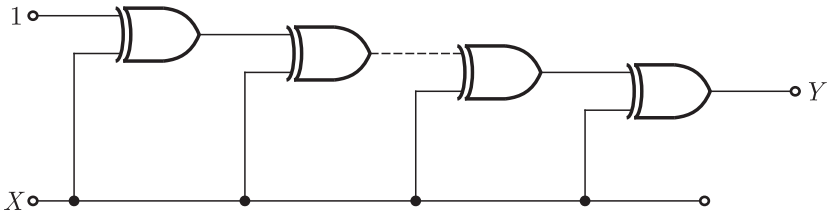
(B) $AB+CD$

(C) $\overline{AB+CD}$

(D) $\overline{(A+B)(C+D)}$

**MCQ 51**

If the input to the digital circuit of the below figure consisting of a cascade of 20 XOR gates is $X$, then what is the output $Y$ ?




(A) 0

(B) 1

(C) $X'$

(D) $X$

**MCQ 52**

What are the output bits S (sum) and C (Carry) of a Half Adder having inputs $A = 1$ and $B = 1$ ?

| | S | C |
|---|---|---|
| (A) | 1 | 1 |
| (B) | 1 | 0 |
| (C) | 0 | 1 |
| (D) | 0 | 0 |




**MCQ 53**

The reduced state table of sequential machine has 7 rows. What is the minimum number of flip-flops needed to implement the machine ?

(A) 0

(B) 2

(C) 3

(D) 7

**MCQ 54**

The lower turn off time of MOSFET when compared to BJT can be attributed to which one of the following ?

(A) Input impedance.

(B) Positive temperature coefficient.

(C) Absence of minority carriers.

(D) On-state resistance.

**MCQ 55**

The shunt type regulator is suitable for which of the following ?

(A) Low current, high voltage.

(B) Low current, low voltage.

(C) High current, low voltage.

(D) High current, high voltage.

## YEAR 2006

### MCQ 56

Match **List I** (Device) with **List II** (Application) and select the correct answer using the code given below the lists :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Diode | 1. | Amplifier |
| B. | Transistor | 2. | Oscillator |
| C. | Tunnel diode | 3. | Rectifier |
| D. | Zener diode | 4. | Voltage Regulator |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 4 | 1 | 2 | 3 |
| (B) | 3 | 2 | 1 | 4 |
| (C) | 4 | 2 | 1 | 3 |
| (D) | 3 | 1 | 2 | 4 |

gate help

### MCQ 57

Match **List I** (Type of Amplifier/Configuration) with **List II** (Characteristic Property) and select the correct answer using the code given below the lists :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Common emitter amplifier | 1. | Very low output resistance |
| B. | Emitter follower | 2. | Current gain $\simeq 1$ |
| C. | Common base amplifier | 3. | Beta multiplication |
| D. | Darlington pair | 4. | Very high power gain |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 4 | 1 | 2 | 3 |
| (B) | 2 | 3 | 4 | 1 |
| (C) | 4 | 3 | 2 | 1 |
| (D) | 2 | 1 | 4 | 3 |

**MCQ 58**

Match **List I** (Type of Diode) with **List II** (Characteristics/ Applications) and select the correct answer using the code given below the lists :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Tunnel diode | 1. | Reverse current varies directly with the amount of light |
| B. | Zener diode | 2. | Exhibits negative resistance region in its $I$-$C$ characteristic |
| C. | Photodiode | 3. | Uses only majority carriers and is intended for high frequency operations |
| D. | Schottky diode | 4. | Silicon $p$-$n$ junction diode that is designed for limiting the voltage across the terminals in reverse bias |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 2 | 3 | 1 | 4 |
| (B) | 1 | 4 | 2 | 3 |
| (C) | 2 | 4 | 1 | 3 |
| (D) | 1 | 3 | 2 | 4 |

**MCQ 59**

Match **List I** (Parameter) with **List II** (Variation) and select the correct answer using the code given below the lists :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Electron mobility around room temperature | 1. | Increases with temperature |
| B. | Energy gap | 2. | Decreases with temperature |
| C. | Intrinsic carrier concentration | 3. | Remain constant as temperature is varied |
| D. | Mole density(gm/mole) | | |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|

(A) 2 1 1 1

(B) 1 2 1 3

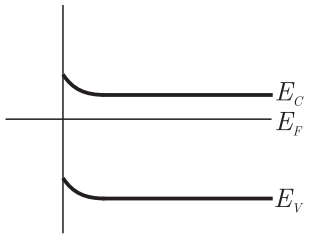
(C) 2 2 1 3

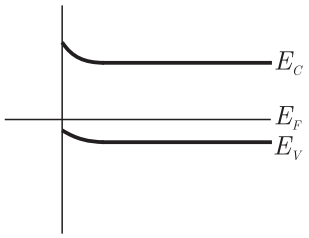
(D) 2 2 1 1

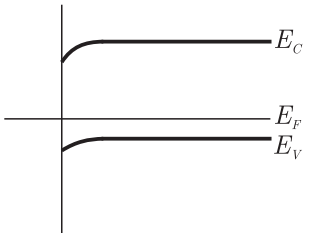
**MCQ 60**

Match **List I** (Metal-semiconductor Band diagram under Equilibrium) with **List II** (Type of contact) and select the correct answer using the code given below the lists :

**List I**

A. 


B. 

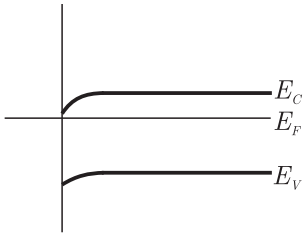

C. 


**List II**

1. Rectifier to $n$ - type

2. Ohmic to $n$ - type

3. Rectifier to $p$ - type

D. 


4. Ohmic to $p$ - type

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 1 | 4 | 2 | 3 |
| (B) | 1 | 4 | 3 | 2 |
| (C) | 4 | 1 | 3 | 2 |
| (D) | 4 | 1 | 2 | 3 |

gate help

**MCQ 61**

Consider the following statements :

Data that are stored at a given address in a random access memory are lost

1. when power goes off.
2. when the data are read from the address.
3. when new data are written at the address.
4. because it is non-volatile memory.

Which of the statements given above are correct ?

(A) 1 and 2

(B) 1, 2 and 4

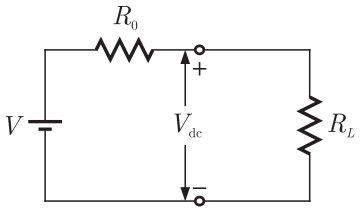
(C) 2 and 3

(D) 1 and 3

**MCQ 62**

The circuit shown below is the Thevenin's equivalent circuit of a centre-tapped full wave rectifier with diode forward resistance $R_f = 100\,\Omega$, transformer secondary coil resistance $R_S = 30\,\Omega$, peak input voltage $V_m = 10\,\text{V}$.

What are the values of $V$ and $R_0$, respectively ?




(A) 10 V, 100 Ω

(B) 6.36 V, 130 Ω

(C) 6.36 V, 115 Ω

(D) 4.54 V, 130 Ω

**MCQ 63**

Which one of the following equations represents the energy gap $(E_G)$ variation of silicon with temperature $(T)$ ?

(A) $E_G(T) = 2.11 - 3.60 \times 10^{-4}\,\mathrm{T}$

(B) $E_G(T) = 1.21 - 3.60 \times 10^{-4}\,\mathrm{T}$

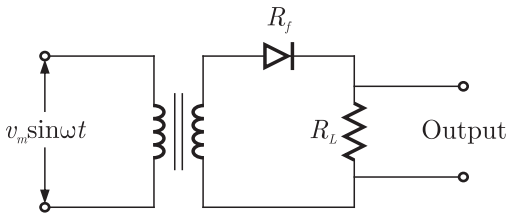
(C) $E_G(T) = 1.41 - 2.23 \times 10^{-4}\,\mathrm{T}$

(D) $E_G(T) = 0.785 - 2.23 \times 10^{-4}\,\mathrm{T}$

**MCQ 64**

Consider the circuit given below where $R_f$ is the diode forward resistance and $R_L$ the load resistance.

What is the average rectified current equal to ?




(A) $V_m/(R_L + R_f)$

(B) $V_m/\{\pi(R_f + R_L)\}$

(C) $2V_m/\pi$

(D) $V_m/\{\sqrt{2}(R_L + R_f)\}$

**MCQ 65**

In a Hall effect experiment, a $p$-type semiconductor sample with hole

concentration $p_1$ is used. The measured value of the Hall voltage is $V_{H1}$. If the $p$-type sample is now replaced by another $p$-type sample with hole concentration $p_2$ where $p_2 = 2p_1$, what is the new Hall voltage $V_{H2}$ ?

(A) $2\,V_{H1}$

(B) $4\,V_{H1}$

(C) $(1/2)\,V_{H1}$

(D) $(1/4)\,V_{H1}$

**MCQ 66**

What is the thermal runaway in a bipolar junction transistor biased in the active region due to ?

(A) Heating of the transistor emitter region.

(B) Changes in 'β' which increases with temperature.

(C) Base emitter voltage $V_{\text{BE}}$ which decreases with rise in temperature.

(D) Increase in reverse collector-base saturation current due to rise in internal device temperature.

**MCQ 67**

The reverse saturation current of a Si-based $p$-$n$ junction diode increases 32 times due to a rise in ambient temperature. If the original temperature was 40°C, what is the final temperature ?

(A) 90°C

(B) 72°C

(C) 45°C

(D) 50°C

**MCQ 68**

What is the main difference between MOSFETs and BJTs in terms of their $I$-$V$ characteristics ?

(A) Current is quadratic with $V_{\text{GS}}$ for MOSFETs and linear with $V_{\text{BE}}$ for BJTs.

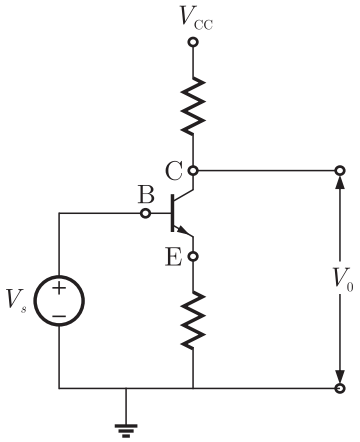
(B) Current is linear with $V_{\text{GS}}$ for MOSFETs and exponential with $V_{\text{BE}}$ for BJTs.

(C) Current is exponential with $V_{\text{GS}}/\,V_{\text{BE}}$ in both these devices, but rise is faster in MOSFETs.

(D) Current is quadratic with $V_{GS}$ for MOSFETs and exponential with $V_{BE}$ for BJTs.

**MCQ 69**

In the circuit given below, if the output is taken from point E instead of node C, what will be the result ?




(A) An increase in the output impedance.

(B) A reduction in the output impedance.

(C) An increase in the input impedance.

(D) A reduction in the input impedance.

**MCQ 70**

If an input periodic signal with non-zero d.c. component is impressed upon a high-pass $RC$ circuit, what will be the d.c. component in the output waveform ?

(A) Zero.

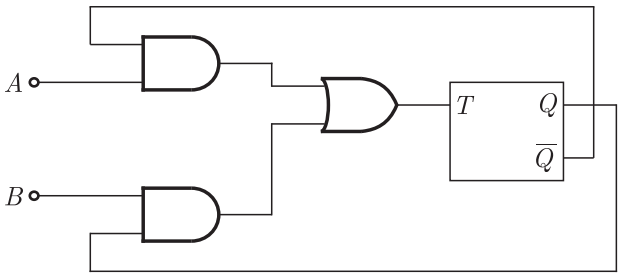
(B) It depends on the value of the capacitor.

(C) It depends on the value of the resistor.

(D) Same as that in input.

**MCQ 71**

What is represented by the digital circuit given below ?




(A) An SR flip-flop with $A = S$ and $B = R$

(B) A JK flip-flop with $A = K$ and $B = J$

(C) A JK flip-flop with $A = J$ and $B = K$

(D) An SR flip-flop with $A = R$ and $B = S$

**MCQ 72**

Which one of the following is not a characteristic of CMOS configuration ?

(A) CMOS devices dissipate much lower static power than bipolar devices.

(B) CMOS devices have low input impedances.

(C) CMOS devices have higher noise margins.

(D) CMOS devices have much lower trans-conductance than bipolar devices.

**MCQ 73**

For an $n$-type semiconductor having any doping level, which of the following hold(s) good :

1. $p_n N_D = n_i^2$
2. $p_p N_D = n_i^2$
3. $n_n N_D = n_i^2$
4. $p_n n_n = n_i^2$

Select the correct answer using the code given below :

(A) 1 and 4

(B) 2 and 4

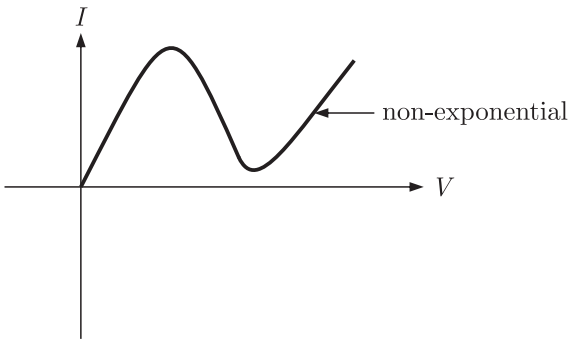
(C) 3 and 4

(D) Only 4

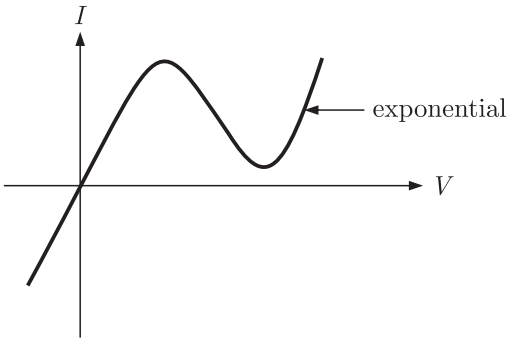
**MCQ 74**

Match the $I$-$V$ characteristics given below with Gunn diode, Photo diode and Tunnel diode.

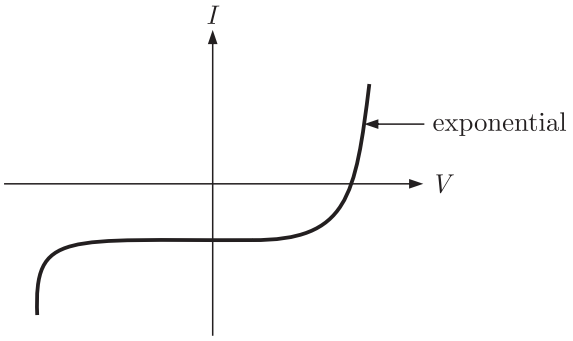
(A)




(B)




(C)




Select the correct answer using the code given below :

| | Gunn diode | Photo diode | Tunnel diode |
|---|---|---|---|
| (A) | A | B | C |
| (B) | B | A | C |
| (C) | A | C | B |
| (D) | B | C | A |

**MCQ 75**

Consider the following statements :

In frequency response of $n$-stage amplifiers,

1. upper cut-off frequency is $(2^{1/n} - 1)$ times that of a single stage.
2. lower cut-off frequency is $(2^{1/n} - 1)$ times that of a single stage.
3. lower cut-off frequency is $(2^{1/n} - 1)^{-1}$ times that of a single stage.
4. upper cut-off frequency is $(2^{1/n} - 1)^{-1}$ times that of a single stage.

Which of the statements given above are correct ?

(A) 1 and 2

(B) 1 and 3

(C) 2 and 4

(D) 3 and 4




**MCQ 76**

Consider the following statements :

An applied bias voltage in a $p$-$n$ junction diode ($n$ region positive with respect to $p$ region) results in

1. increase in potential barrier.
2. reduction in space charge layer width.
3. increase in space charge layer width.
4. increase in magnitude of electric field.

Which of the statements given above are correct ?

(A) 1 and 2

(B) 1 and 3

(C) 1 and 4

(D) 1, 3 and 4

**MCQ 77**

Which one of the following statements is correct in respect of BJT ?

(A) Avalanche multiplication starts when the reverse biased collector-base voltage $V_{CB}$ equals the avalanche breakdown voltage $BV_{CBO}$.

(B) The early effect starts as soon as punch-throgh occurs in a transistor.

(C) The small signal current gain $h_{fe}$ = large signal current gain $h_{FE}$

when $\partial\, h_{\mathrm{FE}} / \partial\, I_C = 0$.

(D) In the CE mode, a transistor can be cut off by reducing $I_B$ to zero.

**MCQ 78**

In a biased step-graded $p$-$n$ junction, what is the correct expression for the equilibrium contact potential ($V_0$) ?

(A) $V_0 = V_T \ln(N_A / N_D n_i^2)$

(B) $V_0 = V_T \ln(N_D / N_A n_i^2)$

(C) $V_0 = V_T \ln(N_A N_D / n_i^2)$

(D) $V_0 = V_T \ln(n_i^2 / N_A N_D)$

where $V_T \propto T/q$, $T$ being the temperature, $q$ the electronic charge, $N_A$ and $N_D$ are the doping levels of the $p$ and $n$ regions, respectively, and $n_i$ is the intrinsic carrier concentration.

**MCQ 79**

In an unbiased $p$-$n$ junction, the junction current at equilibrium is

(A) due to diffusion of majority carriers only.

(B) due to diffusion of minority carriers only.

(C) zero, because equal and opposite drift and diffusion currents for electrons and holes cross the junction.

(D) zero, because no charges cross the junction.

**MCQ 80**

What is the purpose of impedance matching between the output of previous stage and input of next stage and input of next stage in a cascaded amplifier ?

(A) To achieve high efficiency.

(B) To achieve maximum power transfer.

(C) To achieve reduced distortion.

(D) To achieve reduced noise.

**MCQ 81**

In a bridge a.c. to d.c. converter using $p$-$n$ diodes, if the input voltage is $v \sin \omega t$, what is the peak inverse voltage across any diode ?

(A) $v$

(B) $2\,v$

(C) $v/2$

(D) $v/\sqrt{2}$

**MCQ 82**

Why is an external pass resistor used in a voltage regulator ?

(A) For short circuit protection.

(B) For increasing the current that regulator can handle.

(C) For increasing the output voltage.

(D) For improving the regulation.

**Direction :** The following two items consist of two statements, one labelled as Assertion 'A' and the other labelled as Reason 'R'. You are to examine these two statements carefully and decide if the Assertion A and the Reason R are individually true and if so, whether the Reason is a correct explanation of the Assertion. Select your answers to these items using the codes given below.




**Codes :**

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

**MCQ 83**

**Assertion (A) :** In JFET, the phenomenon of thermal runaway is not observed around room temperature.

**Reason (R) :** The heat dissipation in the semiconductor increases its temperature which increase the carrier mobility with temperature.

**MCQ 84**

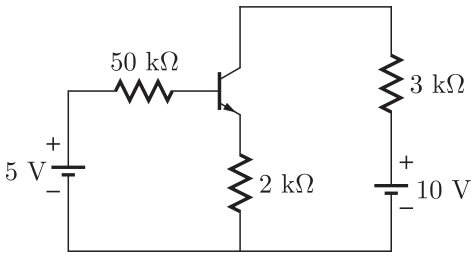
**Assertion (A) :** In a good power supply, the percentage of voltage regulation should be close to zero.

**Reason (R) :** Zero percentage regulation means that there will be no change in output voltage if the load resistance varies between the

limits 0 to $\infty$.

## YEAR 2005

### MCQ 85




For the circuit shown in figure given above, assume $\beta = h_{FE} = 100$. The transistor is in

(A) Active region and $V_{CE} = 5$ V

(B) Saturation region

(C) Active region and $V_{CE} = 1.42$ V

(D) Cut-off region

### MCQ 86

Two $p$-$n$ junction diodes are connected back to back to make a transistor. Which one of the following is correct ?

(A) The current gain of such a transistor will be high.

(B) The current gain of such a transistor will be moderate.

(C) It cannot be used as a transistor due to large base width.

(D) It can be used only for $pnp$ transistor.

### MCQ 87

Which of the following notations have two representations of zero ?

1. 1's complement with radix of number being 2.
2. 7's complement with radix of number being 8.
3. 9's complement with radix of number being 10.
4. 10's complement with radix of number being 10.

Select the correct answer using the codes given below :

(A) 1, 2 and 4

(B) 1 and 3

(C) 2, 3 and 4

(D) 1, 2 and 3

**MCQ 88**

In a $p$-type silicon sample, the hole concentration is $2.25 \times 10^{15}$/cc . If the intrinsic carrier concentration is $1.5 \times 10^{10}$/cc, what is the electron concentration in the $p$-type silicon sample

(A) zero

(B) $10^{10}$/cc

(C) $10^{5}$/cc

(D) $1.5 \times 10^{25}$/cc

**MCQ 89**

What is the reverse recovery time of a diode when switched from forward bias $V_F$ to reverse bias $V_R$ ?

(A) Time taken to remove the stored minority carriers.

(B) Time taken by the diode voltage to attain zero value.

(C) Time to remove stored minority carriers plus the time to bring the diode voltage to reverse bias $V_R$.

(D) Time taken by the diode current to reverse.

**MCQ 90**

Which one of the following statements is correct ?

In a transistor,

(A) $I_{\text{CBO}}$ is greater than $I_{\text{CEO}}$, and does not depend upon temperature.

(B) $I_{\text{CBO}}$ is greater than $I_{\text{CO}}$, and doubles for every ten degrees rise in temperature.

(C) $I_{\text{CBO}}$ is equal to $I_{\text{CO}}$ and double for every ten degrees rise in temperature.

(D) $I_{\text{CEO}}$ is equal to $I_{\text{CO}}$ and doubles for every ten degrees rise in temperature.

**MCQ 91**

Consider following statements :

1. BJT is a current controlled device with high input impedance and high gain bandwidth.
2. FET is a voltage controlled device with high input impedance and low gain bandwidth.
3. UJT is a negative resistance device and can be used as an oscillator.
4. BJT, FET and UJT can all be used for amplification.

Which of the statements given above are correct ?

(A) 1 and 2

(B) 2 and 3

(C) 3 and 4

(D) 1 and 4

**MCQ 92**

The dynamic transfer characteristics of a transistor is represented by

$$I_C = A_1 I_B + A_2 I_B^2$$

Where $A_1$ and $A_2$ are constants. If input signal $I_B = I_1 \cos \omega_1 t + I_2 \cos \omega_2 t$ the output will contain

(A) $\omega_1, \omega_2, 2\omega_1, 2\omega_2$

(B) dc term, $\omega_1, \omega_2, \omega_1 + \omega_2, \omega_1 - \omega_2$

(C) dc term, $\omega_1, \omega_2, 2\omega_1, 2\omega_2, 2\omega_1 + 2\omega_2, 2\omega_1 - 2\omega_2$

(D) dc term, $\omega_1, \omega_2, 2\omega_1, 2\omega_2, \omega_1 + \omega_2, \omega_1 - \omega_2$

**MCQ 93**

Find the break region (voltage range) over which the dynamic resistance of a diode is multiplied by a factor of 1000. Let this region be contained between $v_1$ and $v_2$, then then is $|v_1 - v_2|$ given by

(A) $\log_C (1000\, V_T)$

(B) $1000\, V_T$

(C) $(\log_e 10^3)\, V_T$

(D) The value cannot be computed with the given data

**MCQ 94**

An emitter in a bipolar junction transistor is doped much more heavily than the base as it increases the

(A) Emitter efficiency.

(B) Base transport factor.

(C) Forward current gain.

(D) All the three given above.

**MCQ 95**

Match **List I** (Type of Device) with **List II** (Characteristics/ Application) and select the correct answer using the codes given below the lists :

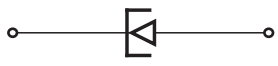
| | **List I** | | **List II** |
|---|---|---|---|
| A. | Zener diode | 1. | Display panel |
| B. | Tunnel diode | 2. | Voltage reference |
| C. | Schottky diode | 3. | Light detection |
| D. | Photo diode | 4. | Negative resistance |
| | | 5. | High frequency switching |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 3 | 4 | 5 | 2 |
| (B) | 2 | 5 | 1 | 3 |
| (C) | 3 | 5 | 1 | 2 |
| (D) | 2 | 4 | 5 | 3 |

**MCQ 96**

Match **List I** (Circuit Symbol) with **List II** (Device) and select the correct answer using the code given below the lists :

| **List I** | | **List II** |
|---|---|---|
| A. |  | 1. N-channel FET |
| B. |  | 2. Varactor |

C.  3. Tunnel diode

D.  4. P-channel MOSFET

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 3 | 2 | 1 | 4 |
| (B) | 1 | 4 | 3 | 2 |
| (C) | 3 | 4 | 1 | 2 |
| (D) | 1 | 2 | 3 | 4 |

### MCQ 97

Match **List I** (Device) with **List II** (Application) and select the correct answer using the code given below the lists :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | *p-n* junction diode | 1. | Microwave generator |
| B. | tunnel diode | 2. | Low frequency rectifier |
| C. | JFET | 3. | High frequency rectifier |
| D. | Schottky barrier diode | 4. | Voltage variable resistor |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 2 | 3 | 4 | 1 |
| (B) | 4 | 1 | 2 | 3 |
| (C) | 2 | 1 | 4 | 3 |
| (D) | 4 | 3 | 2 | 1 |

### MCQ 98

Consider the following statements :

1. A Hartley oscillator circuit uses a tapped inductor for inductive feedback.

2. Oscillator circuit can be operated in class A condition for better wave shape.
3. Frequency stabilization is obtained by use of automatic biasing.

Which of the statements given above are correct ?

(A) 1, 2 and 3

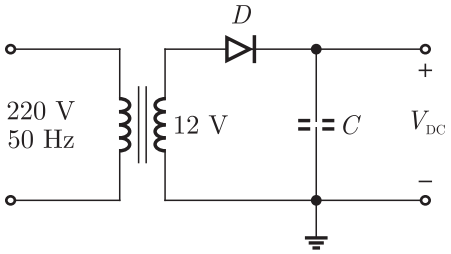
(B) 1 and 2

(C) 2 and 3

(D) 1 and 3

**MCQ 99**

In the rectifier circuit shown above, what should be minimum peak-inverse-voltage (PIV) rating of the diode ?




(A) 12 V

(B) 12 $\sqrt{2}$ V

(C) 24 V

(D) $24\sqrt{2}$ V

**MCQ 100**

Which one of the following statements is correct ? In the case of load regulation

(A) when the temperature changes, the output voltage remains constant.

(B) when the input voltage changes, the load current remains constant.

(C) when the load changes, the load current remains constant.

(D) when the load changes, the output voltage remains constant.

**MCQ 101**

In a centre tap full wave rectifier, 100 V is the peak voltage between the centre tap and one end of the secondary. What is the maximum voltage across the reverse biased diode ?

(A) 200 V

(B) 141 V

(C) 100 V

(D) 86 V

### MCQ 102

Match **List I** (Type of Logic) with **List II** (Characteristics/ Application) and select the correct answer using the code given below the lists :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Direct coupled logic | 1. | Good for monolithic I.C. |
| B. | Diode transistor logic | 2. | Slow speed of operation |
| C. | Emitter coupled logic | 3. | Very fast speed of operation |
| D. | Resistor transistor logic | 4. | Current hogging |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 2 | 1 | 3 | 4 |
| (B) | 4 | 3 | 1 | 2 |
| (C) | 2 | 3 | 1 | 4 |
| (D) | 4 | 1 | 3 | 2 |

### MCQ 103

Which of the following statements is not correct ?

(A) $X + \overline{X}Y = X$

(B) $X(\overline{X} + Y) = XY$

(C) $X + X\overline{Y} = X$

(D) $ZX + Z\overline{X}Y = ZX + ZY$

### MCQ 104

The Boolean expression $\overline{Y}\,\overline{Z} + \overline{X}\,\overline{Z} + \overline{X}\,\overline{Y}$ is logically equivalent to

(A) $Y\,Z + \overline{X}$

(B) $Y\,Z\,X+\overline{X}\,\overline{Y}\,\overline{Z}$

(C) $Y\,Z+X\,Z+X\,Y$

(D) $X\,\overline{Y}\,\overline{Z}+\overline{X}\,\overline{Y}\,\overline{Z}+\overline{X}\,Y\,\overline{Z}+\overline{X}\,\overline{Y}\,Z$

**MCQ 105**

Which one of the following statements is correct ? Removing the small resistance ($\sim 100\,\Omega$) in the collector lead of the pull-up transistor of a totempole output gate, will result in

(A) reduced switching time from $V_{\text{out}}(1)$ to $V_{\text{out}}(0)$.

(B) incorrect operation of the gate.

(C) lower power dissipation.

(D) more noise generation in the power supply distribution at high frequency.




**MCQ 106**

In a ripple counter, the state whose output has a frequency equal to $1/8^{\text{th}}$ that of the clock signal applied to the first stage, also has an output periodicity equal to $1/8^{\text{th}}$ that of the output signal obtained from the last stage. The counter is

(A) Modulo - 8

(B) Modulo - 6

(C) Modulo - 64

(D) Modulo -16

**MCQ 107**

Consider the following statements :

In amplifiers,

1. a complementary symmetry amplifier has 1 *pnp* and 1 *npn* transistor.
2. a boot strap incorporates emitter follower.
3. the main function of transformer used in the output of a power amplifier is to increase its voltage gain.
4. the harmonic distortion of the signal produced in a $RC$ coupled transistor amplifier is due to transformer itself.

Which of the statements given above are correct ?

(A) 1, 2 and 3

(B) 2, 3 and 4

(C) 1, 3 and 4

(D) 1, 2 and 4

**MCQ 108**

In a single stage $RC$-coupled amplifier state, what are the phase shifts introduced at lower and upper 3-dB frequencies, respectively ?

(A) $45^\circ, 225^\circ$

(B) $45^\circ, 135^\circ$

(C) $90^\circ, 180^\circ$

(D) $45^\circ, 180^\circ$

**MCQ 109**

For common emitter configuration which one of the following statements is correct ?

(A) large current gain and high input resistance.

(B) large voltage gain and low output resistance.

(C) small voltage gain and low input resistance.

(D) small current gain and high output resistance.

**MCQ 110**

The gain of a bipolar transistor drops at high frequencies. This is because of the

(A) Coupling and bypass capacitors.

(B) Early effect.

(C) Inter-electrode transistor capacitances

(D) Coupling and bypass capacitors, and inter-electrode transistor capacitances

**Direction :** The following two items consist of two statements, one labelled as Assertion 'A' and the other labelled as Reason 'R'. You are to examine these two statements carefully and decide if the Assertion A and the Reason R are individually true and if so, whether the Reason is a correct explanation of the Assertion.Select your answers to these items using the codes give below.

**Codes :**

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

### MCQ 111

**Assertion (A) :** Emitter coupled oscillator is capable of high frequency operation.

**Reason (R) :** It consists of saturating *npn* BJTs.

### MCQ 112

**Assertion (A) :** An FET operated crystal oscillator operates on the concept of feedback.

**Reason (R) :** The feedback is provided by the drain-to-gate capacitance $C_{dg}$.




## YEAR 2004

### MCQ 113

Two identical $RC$ coupled amplifiers, each having an upper cut-off frequency $f_u$, are cascaded with negligible loading. What is the upper cut-off frequency of the overall amplifier ?

(A) $\dfrac{f_u}{\sqrt{\sqrt{2}-1}}$

(B) $f_u\sqrt{\sqrt{2}-1}$

(C) $f_u/2$

(D) $2f_u$

### MCQ 114

Two identical $RC$ coupled amplifiers, each having a lower cut-off frequency $f_l$, are cascaded with negligible loading. What is the lower cut-off frequency of the overall amplifier ?

(A) $\dfrac{f_l}{\sqrt{\sqrt{2}-1}}$

(B) $f_l\sqrt{\sqrt{2}-1}$

(C) $f_l/2$

(D) $2f_l$

**MCQ 115**

A synchronous sequential circuit is designed to detect a bit sequence 0101 (overlapping sequence included). Everytime this sequence is detected, the circuit produces an output of '1'. What is the minimum number of states the circuit must have ?

(A) 4

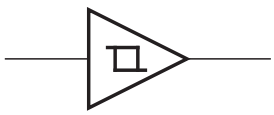
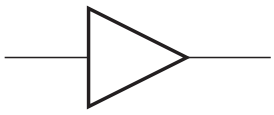
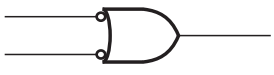
(B) 5

(C) 6

(D) 7

**MCQ 116**

Match **List I** (Circuit Symbols) with **List II** (Nomenclature) and select the correct answer using the codes given below :

| **List I** | | **List II** |
|---|---|---|
| A. |  | 1. NAND |
| B. |  | 2. NOR |
| C. |  | 3. Buffer |
| D. |  | 4. Schmitt trigger |

**Codes :**

| A | B | C | D |
|---|---|---|---|

(A) 4 3 1 2

(B) 3 4 2 1

(C) 4 3 2 1

(D) 3 4 1 2

### MCQ 117

If $x$ and $y$ are Boolean variables, which one of the following is the equivalent of $x \oplus y \oplus xy$ ?

(A) $x + \overline{y}$

(B) $x + y$

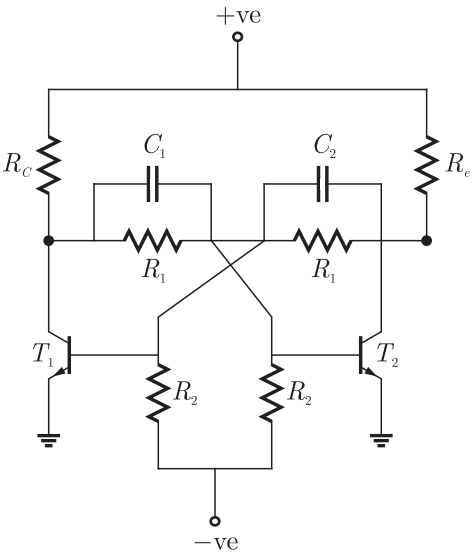
(C) 0

(D) 1




### MCQ 118

Consider the following bistable multivibrator circuit ?




Which one of the following statements is correct ?

In the above circuit, condensers $C_1$ and $C_2$ are used

(A) for passing the noise

(B) to compensate the collector-emitter capacitance

(C) to speed up the switching action

(D) to provide negative feedback path

**MCQ 119**

Consider the following statements :

1. TTL has high switching speed and good fan-out capability.
2. ECL has the least propagation delay.
3. $I^2L$ uses multi-collector transistors.
4. NMOS has more silicon area.

Which of the statements given above are correct ?

(A) 1, 2 and 3

(B) 2 and 4

(C) 1, 3 and 4

(D) 1, 2, 3 and 4

**MCQ 120**

Which one of the following statements is correct ?

An ideal regulated power supply should have

(A) 100% regulation

(B) 50% regulation

(C) 0% regulation

(D) 75% regulation




**MCQ 121**

Match **List I** (operation) with **List II** (Associated Device) and select the correct answer using the codes given below :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Counting | 1. | ROM |
| B. | Decoding | 2. | Multiplexer |
| C. | Data selection | 3. | Demultiplexer |
| D. | Code conversion | 4. | Register |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 3 | 4 | 2 | 1 |
| (B) | 3 | 4 | 1 | 2 |
| (C) | 4 | 3 | 1 | 2 |

(D) 4 3 2 1

**MCQ 122**

Which one of the following statements is correct ?

A photodiode works on the principle of

(A) photo-voltaic effect

(B) photo-conductive effect

(C) photo-electric effect

(D) photo-thermal effect

**MCQ 123**

Match **List I** (Circuit) with **List II** (Application) and select the correct answer using the codes given below :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Monostable multivibrator | 1. | Comparator |
| B. | Bistable multivibrator translator | 2. | d.c. level |
| C. | Clamping circuit | 3. | Delay |
| D. | Schmitt trigger | 4. | Voltage controlled oscillator |
| | | 5. | Counter |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 4 | 3 | 1 | 2 |
| (B) | 3 | 5 | 2 | 1 |
| (C) | 4 | 5 | 2 | 1 |
| (D) | 5 | 4 | 1 | 2 |

**MCQ 124**

What are the values respectively, of $R_1$ and $R_2$ in the expression

$$(235)_{R_1} = (565)_{10} = (1065)_{R_2} \text{ ?}$$

(A) 8, 16

(B) 16, 8

(C) 6, 16

(D) 12, 8

**MCQ 125**

Match **List I** (Sections of a Service Voltage Regulator) with **List II** (Elements used in these Sections) and select the correct answer using the codes given below :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Reference source | 1. | Op-amp |
| B. | Error detector | 2. | BJT |
| C. | Control device | 3. | Zener diode |
| D. | Current limit | 4. | Short circuit protection |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 3 | 1 | 2 | 4 |
| (B) | 3 | 2 | 1 | 4 |
| (C) | 4 | 2 | 1 | 3 |
| (D) | 4 | 1 | 2 | 3 |




**MCQ 126**

Which one of the following statements is correct ?

The efficiency of class B push-pull amplifiers is much higher than that of class A amplifiers primarily because

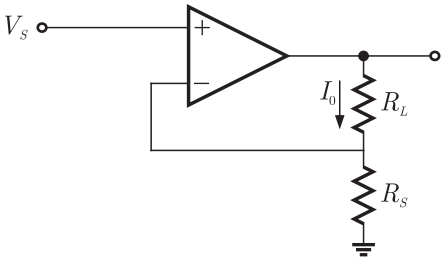
(A) the distortion is kept within acceptable limits

(B) one half of the input signal is amplified using one transistor and the other half is phase-inverted and fed to the other transistor

(C) matched pair transistors are used in the class B push-pull operation

(D) the quiescent d.c. current is avoided

**MCQ 127**




Which one of the following is the correct expression for the current $I_0$ ?

(A) $\frac{V_S.R_L}{R_S(R_L+R_S)}$

(B) $\frac{V_S}{R_S}$

(C) $\frac{V_S}{R_L}$

(D) $V_S\left(\frac{1}{R_L}+\frac{1}{R_S}\right)$

gate help

**MCQ 128**

It is required to construct a $2^n$-to-1 multiplexer by using 2-to-1 multiplexers only. How many of 2-to-1 multiplexer are needed ?

(A) $n$

(B) $2^{2n}$

(C) $2^{n-1}$

(D) $2^n - 1$

**MCQ 129**

A inverter gate has guaranteed output levels as : logic '1' = 3.8 V and logic '0' = 0.7 V. The maximum low level input voltage at which the output remains high = 2 V. The minimum high-level input voltage at which the output remains low = 3.1 V. What are the noise margins of this gate ?

(A) $NM_H = 2.4\,\text{V}, NM_L = 1.8\,\text{V}$

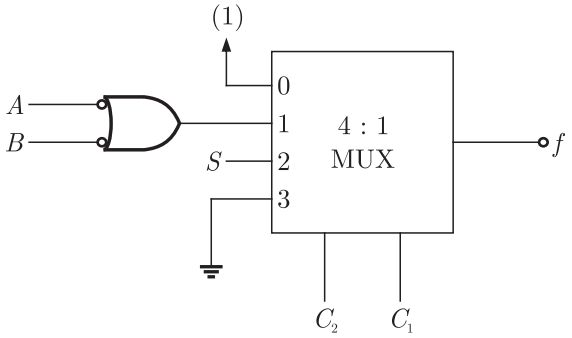
(B) $NM_H = 1.8\,\text{V}, NM_L = 1.3\,\text{V}$

(C) $NM_H = 0.7\,\text{V}, NM_L = 1.8\,\text{V}$

(D) $NM_H = 0.7\,\text{V}, NM_L = 1.3\,\text{V}$

**MCQ 130**

Consider the following circuit :




Which one of the following gives the function implemented by the MUX-based digital circuit ?

(A) $f = C_2.\overline{C_1}.S + \overline{C_2}.C_1.(\overline{A} + \overline{B})$

(B) $f = \overline{C_2}.\overline{C_1} + C_2.C_1 + C_2.\overline{C_1}.S + \overline{C_2}.C_1.\overline{AB}$

(C) $f = \overline{AB} + S$

(D) $f = \overline{C_2}.\overline{C_1} + C_2.\overline{C_1}.S + \overline{C_2}.C_1.\overline{AB}$

**MCQ 131**

Which one of the following statements is correct ?

For a 4-input NOR gate, when only two inputs are to be used, the best option for the unused inputs is to

(A) connect them to the ground

(B) connect them to $V_{CC}$

(C) keep them open

(D) connect them to the used inputs

**MCQ 132**

A range decoder is a digital circuit which outputs a '1' whenever an m-bit number $X$ falls within the range, $2^p \leq X \leq 2^q, 0 \leq p, q \leq m-1$

Which one of the following functions describes the range-decoder ?

(A) $(X_p.X_{p+1}.---.X_{q-1}.X_q)$
$(X_{q+1} + X_{q+2} + --- + X_{m-1})$

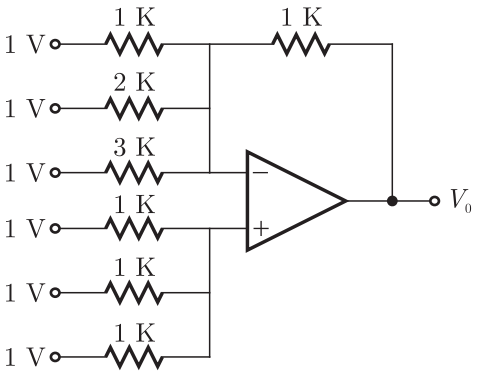
(B) $(X_p \oplus X_{p+1} \oplus --- X_q)$
$(\overline{X}_{q+1} + \overline{X}_{q+2} + --- + \overline{X}_{m-1})$

(C) $X_p + X_{p+1} + - - - + X_{q-1} + X_q$
$(\overline{X}_{q+1} . - - - . \overline{X}_{m-1})$

(D) $(X_0 + X_1 + X_2 + - - - + X_q)$
$(\overline{X}_{q+1} . \overline{X}_{q+2} . - - - . \overline{X}_{m-1})$

**MCQ 133**

Consider the following circuit :




What is the output voltage $V_0$ in the above circuit ?

(A) 9.5 V

(B) 3 V

(C) 32.2 V

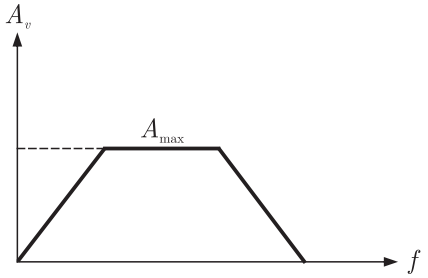
(D) 1 V

**MCQ 134**

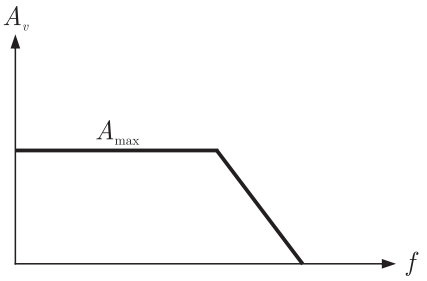
If the coupling capacitors of a CE transistor amplifier is shorted, which one of the following graphs will represent the frequency response curve of the amplifier ?

($A_v$ =voltage gain, $f$= frequency in Hertz, $A_{\max}$ =maximum value of $A_v$)

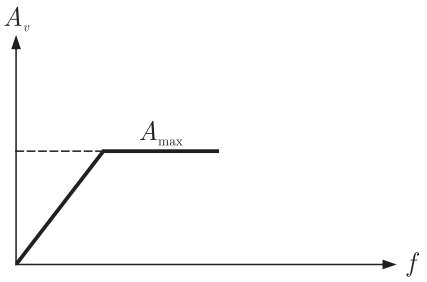
(A)



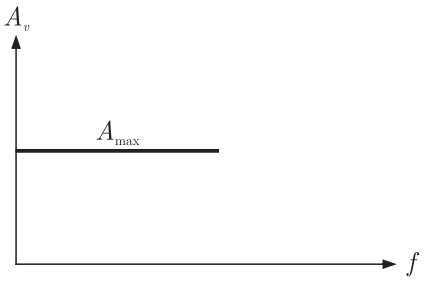
(B)




(C)




(D)



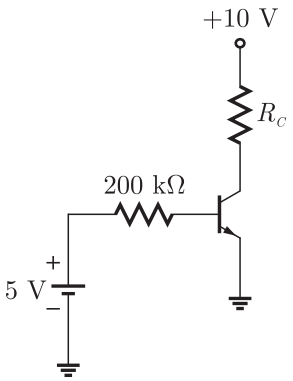

gate help

**MCQ 135**

A silicon transistor with

$$V_{BE_{sat}} = 0.8\text{ V}, \beta_{dc} = 100 \text{ and } V_{CE_{sat}} = 0.2\text{ V}$$

is used in the circuit shown below :



What is the minimum value of $R_C$ for which transistor is in saturation?

(A) $4286\,\Omega$

(B) $4667\,\Omega$

(C) $5000\,\Omega$

(D) $1000\,\Omega$

### MCQ 136

For a JK flip-flop, $Q_n$ is output at time step $t_n$. Which of the following Boolean expressions represents $Q_{n+1}$ ?

(A) $J_n\,\overline{Q}_n + \overline{K}_n\,Q_n$

(B) $J_n\,Q_n + K_n\,\overline{Q}_n$

(C) $\overline{J}_n\,Q_n + K_n\,\overline{Q}_n$

(D) $J_n\,Q_n + \overline{K}_n\,\overline{Q}_n$




### MCQ 137

Consider the following multiplication :

$(10w\,1z)_2 \times (15)_{10} = (y0\,10\,11\,001)_2$

Which one of the following gives appropriate values of $w, y\ and\ z$ ?

(A) $w = 0, y = 0, z = 1$

(B) $w = 0, y = 1, z = 1$

(C) $w = 1, y = 1, z = 1$

(D) $w = 1, y = 1, z = 0$

## YEAR 2003

### MCQ 138

In a $p$-$n$ junction, to make the depletion region extent predominantly into $p$-region, the concentration of impurities in the $p$-region must be

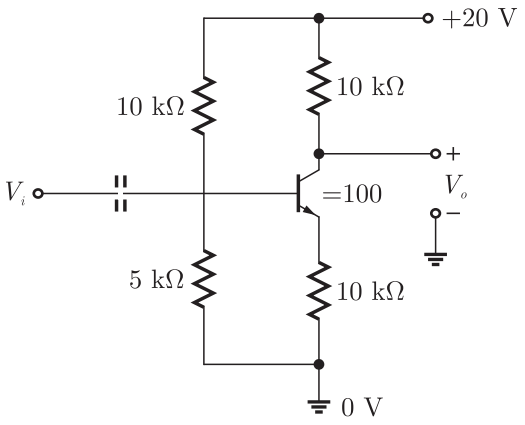
(A) Much less than the concentration of impurities in $n$-region

(B) Much higher than the concentration of impurities in $n$-region

(C) Equal to the concentration of impurities in $n$-region

(D) Zero

### MCQ 139




Consider a silicon transistor connected as a common emitter amplifier as shown above. The quiescent collector voltage of the circuit is approximately

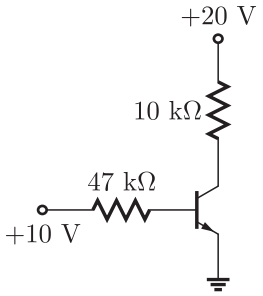
(A) 6.67 V

(B) 10 V

(C) 14 V

(D) 20 V

**MCQ 140**




In the transistor circuit as shown above, the collector to ground voltage is $+20\,\text{V}$. The possible condition is

(A) Collector-emitter terminals shorted

(B) Emitter to ground connection open

(C) 10 kilo-ohms resistor open

(D) Collector-base terminals shorted

**MCQ 141**

In an $RC$ coupled common emitter amplifier

(A) Coupling capacitance affects the high frequency response and bypass capacitance affects the low frequency response

(B) Both coupling and bypass capacitances affect the high frequency response

(C) Both coupling and bypass capacitances affect the low frequency response only

(D) Coupling capacitance affects the low frequency response and the bypass capacitance affects the high frequency response

**MCQ 142**

Consider the following with reference to a CE transistor amplifier :

1. The use of negative feedback
2. The conversion of d.c. power to a.c.
3. High voltage and current gains
4. The use of a step-up transformer

The power gain is due to

(A) 1 and 2

(B) 2 and 3

(C) 1 and 3

(D) 1 and 4

**MCQ 143**

The open-loop voltage gain of an amplifier is 240. The noise level in the output without feedback is 100 mV. If a negative feedback with $\beta = 1/60$ is used, the noise level in the output will be

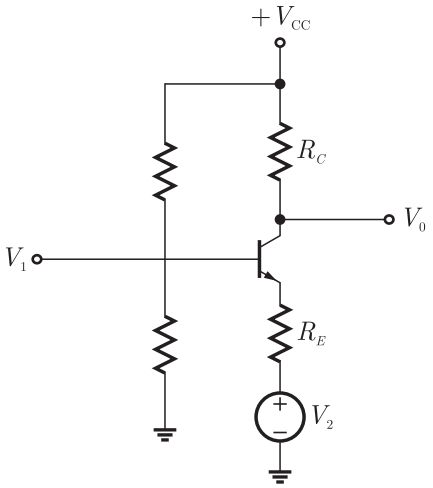
(A) 1.66 mV

(B) 2.4 mV

(C) 4.0 mV

(D) 20 mV

**MCQ 144**




In the circuit as shown above, the ratio of $V_0$ to $(V_2 - V_1)$ would approximately (neglecting constant due to $V_{CC}$) be

(A) $R_C/R_E$

(B) $R_E/R_C$

(C) $-R_C/R_E$

(D) $-R_E/R_C$




**MCQ 145**

An op-amp has a differential gain of $10^3$ and a CMRR of 100. The output voltage of the op-amp with inputs of 120 μV and 80 μV will be

(A) 26 mV

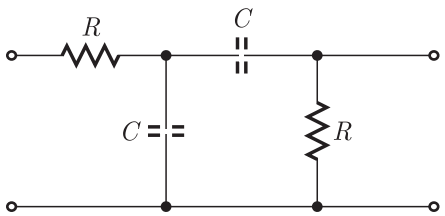
(B) 41 mV

(C) 100 mV

(D) 200 mV

**MCQ 146**

An FET oscillator uses the given phase shift network as shown below. The minimum gain required for oscillation is



(A) $-29$

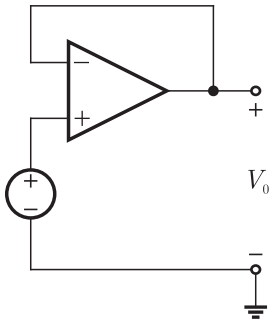
(B) 1

(C) 3

(D) 29

**MCQ 147**

Consider the following statements with reference to an ideal voltage follower circuit as shown below :




gate help

1. Unity gain and no phase shift
2. Infinite gain and 180° phase shift
3. Very high input impedance and very low output impedance
4. It is a buffer amplifier

Which of these statements are correct ?

(A) 1 and 3

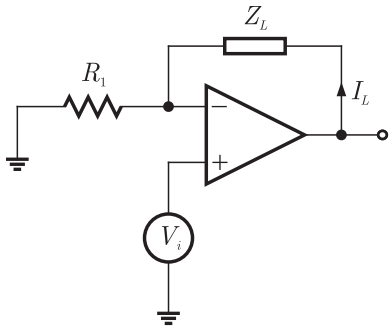
(B) 2 and 4

(C) 1, 3 and 4

(D) 1, 2, 3 and 4

**MCQ 148**

In the op-amp circuit as shown below, the current $I_L$ is




(A) $V_i/Z_L$

(B) $V_i/Z_L \parallel R_1$

(C) $V_i/R_1$

(D) $V_i(R_1 + Z_L)$

**MCQ 149**

Match **List I** (Circuits) with **List II** (Characteristic/Applications) and select the correct answer :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | High-pass $RC$ circuit | 1. | Comparator |
| B. | Low-pass $RC$ circuit | 2. | DC Restorer |
| C. | Clamping circuit | 3. | Integrator |
| D. | Clipping circuit | 4. | Differentiator |
| | | 5. | Compensated Attenuator |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 5 | 4 | 2 | 1 |
| (B) | 4 | 3 | 1 | 2 |
| (C) | 5 | 4 | 1 | 2 |
| (D) | 4 | 3 | 2 | 1 |

**MCQ 150**

The simplified form of a logic function $Y = \overline{(\overline{A\overline{B}}).(\overline{\overline{A}B})}$ is

(A) $A + B$

(B) AB

(C) $\overline{A}+\overline{B}$

(D) $\overline{A}B+A\overline{B}$

**MCQ 151**

The reduced form of the Boolean expression $A[B+C(\overline{AB+AC})]$ is

(A) $\overline{A}B$

(B) $A\overline{B}$

(C) $AB$

(D) $AB+B\overline{C}$

**MCQ 152**

An $n$-bit A/D converter is required to convert analog input in the range 0-5 V to an accuracy of 10 mV. The value of $n$ should be

(A) 16

(B) 10

(C) 9

(D) 8

**MCQ 153**

Consider the following statements in respect of ECL gate :

1. Its switching speed is high
2. It provides OR and NOR logic operations
3. Its power dissipation is small as compared to other logic gates
4. Its logic levels are compatible with other logic family gates

Which of these statements are correct ?

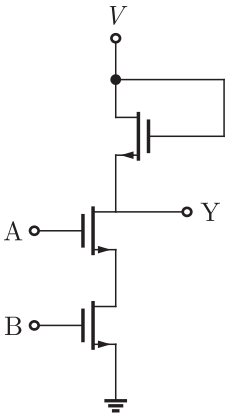
(A) 1 and 2

(B) 1, 2 and 3

(C) 1, 2 and 4

(D) 3 and 4

**MCQ 154**




The above shown NMOS circuit is a gate of the type

(A) NAND

(B) NOR

(C) AND

(D) EXCLUSIVE-OR




**MCQ 155**

D flip-flop can be made from a J-K flip-flop by making

(A) $J = K$

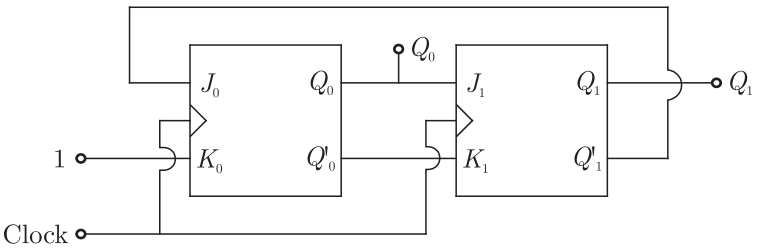
(B) $J = K = 1$

(C) $J = 0, K = 1$

(D) $J = \overline{K}$

**MCQ 156**

In the circuit as shown below, assuming initially $Q_0 = Q_1 = 0$. Then the states of $Q_0$ and $Q_1$ immediately after the 33rd pulse are




(A) 1 1

(B) 1 0

(C) 0 1

(D) 0 0

**MCQ 157**

Which of the following statements is not correct ? Conversion of EXCESS-3 code to BCD can be achieved by using

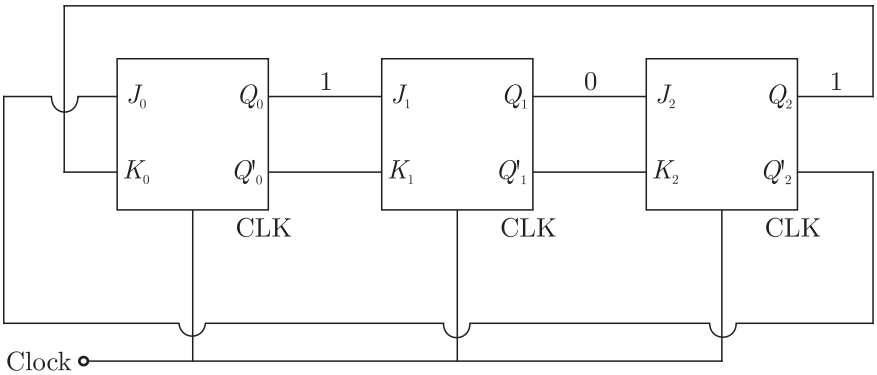
(A) Discrete gates

(B) 4 : 16 de-multiplexer

(C) A 4-bit full adder

(D) A 4-bit half adder

**MCQ 158**




The three-stage Johnson-Ring Counter as shown above is clocked at a constant frequency of $f_c$ from the starting state of $Q_0 Q_1 Q_2 = 1\ 0\ 1$ . The frequency of outputs $Q_0 Q_1 Q_2$ will be

(A) $\frac{f_c}{8}$

(B) $\frac{f_c}{6}$

(C) $\frac{f_c}{3}$

(D) $\frac{f_c}{2}$

**MCQ 159**

A 4-bit modulo-16 ripple counter uses J-K flip-flop. If the propagation delay of each flip-flop is 50 nanoseconds, the maximum clock frequency that can be used is equal to

(A) 20 MHz

(B) 10 MHz

(C) 8 MHz

(D) 5 MHz

### MCQ 160

The binary representation 100110 is numerically equivalent to the

1. Decimal representation 46
2. Octal representation 46
3. Hexadecimal representation 26
4. Excess-3 representation 13

Select the correct answer using the codes given below :

(A) 1 and 2

(B) 2 and 3

(C) 1 and 3

(D) 2 and 4

**Direction :** The following two items consist of two statements, one labelled as Assertion 'A' and the other labelled as Reason 'R'. You are to examine these two statements carefully and decide if the Assertion A and the Reason R are individually true and if so, whether the Reason is a correct explanation of the Assertion. Select your answers to these items using the codes given below.

**Codes :**

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

### MCQ 161

**Assertion (A) :** Transistor $h$-parameter equivalent circuit can be used for the analysis irrespective of the configuration (CE, CB or CC) of the transistor used.

**Reason (R) :** The value of $h$-parameter remain unchanged with transistor configuration.

### MCQ 162

**Assertion (A) :** In an asynchronous counter, output of each FF serves as CLK input signal for the previous FF.

**Reason (R) :** All the FFs do not change states in exact synchronism with the clock pulses.

## YEAR 2002

### MCQ 163

The bonding forces in compound semi-conductors, such as GaAs, arise from

(A) ionic bonding

(B) metallic bonding

(C) covalent bonding

(D) combination of ionic and covalent bonding




### MCQ 164

Consider the following statements in connection with the biasing of semiconductor diodes :

1. LEDs are used under forward-bias condition
2. Photodiodes are used under forward-bias condition
3. Zener diodes are used under reverse-bias condition
4. Variable capacitance diodes are used under reverse-bias condition

Which of these statements are correct ?

(A) 1, 2 and 3

(B) 1, 2 and 4

(C) 2, 3 and 4

(D) 1, 3 and 4

### MCQ 165

The junction capacitance of a linearly graded junction varies with the applied reverse bias $V_r$ as

(A) $V_r^{-1}$

(B) $V_r^{-1/2}$

(C) $V_r^{-1/3}$

(D) $V_r^{1/2}$

**MCQ 166**

The diffusion capacitance of a forward biased $p^+ - n$ junction diode with a steady current $I$ depends on

(A) width of the depletion region

(B) mean lifetime of the holes

(C) mean lifetime of the electrons

(D) junction area

**MCQ 167**

The modified work function of an $n$-channel MOSFET is $-0.85\,\text{V}$. If the interface charge is $3 \times 10^{-4}\,\text{C/m}^2$ and the oxide capacitance is $300\,\mu\text{F/m}^2$, the flat band voltage is

(A) $-1.85\,\text{V}$

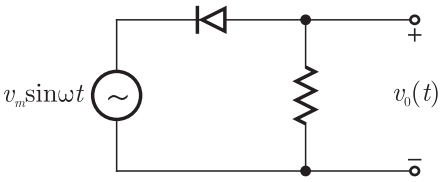
(B) $-0.15\,\text{V}$

(C) $+0.15\,\text{V}$

(D) $+1.85\,\text{V}$

**MCQ 168**

In the circuit shown below, the average value of $v_0(t)$ will be




(A) 0

(B) $-v_m/\pi$

(C) $-v_m/\sqrt{2}$

(D) $-v_m$

**MCQ 169**

Match **List I** (Amplifier's mode of operation) with **List II** (Properties/ characteristics) and select the correct answer using the codes given

below the lists :

| | List I | | List II |
|---|---|---|---|
| A. | Class A | 1. | Clips off half a cycle |
| B. | Class B | 2. | Leads to most stable biasing circuit |
| C. | Class C | 3. | Transistor acts as switch |
| D. | Class D | 4. | Amplification of the resonant frequency only |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 1 | 2 | 3 | 4 |
| (B) | 2 | 1 | 4 | 3 |
| (C) | 1 | 2 | 4 | 3 |
| (D) | 2 | 1 | 3 | 4 |

gate help

### MCQ 170

Early effect in BJT refers to

(A) avalanche breakdown

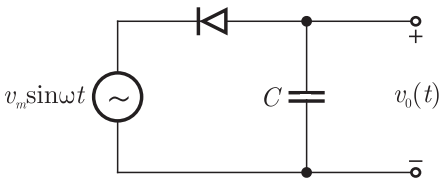
(B) thermal runaway

(C) base narrowing

(D) zener breakdown

### MCQ 171

In the circuit shown below, the average value of $v_0(t)$ will be




(A) 0

(B) $-v_m/\pi$

(C) $-v_m/\sqrt{2}$

(D) $-v_m$

**MCQ 172**

The Darlington pair is mainly used for

(A) impedance matching

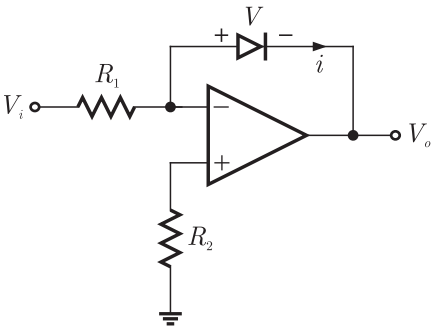
(B) wideband voltage amplification

(C) power amplification

(D) reducing distortion

**MCQ 173**

In the op-amp circuit shown below, $V_i > 0$ and $i = I_0 e^{\alpha V}$. The output $V_o$ will be proportional to




(A) $\sqrt{V_i}$

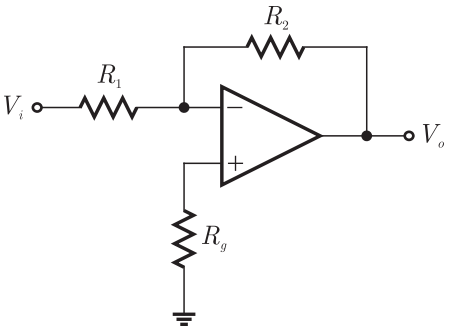
(B) $V_i$

(C) $e^{kV_i}$

(D) $\ln(\mathrm{k}\,V_i)$

**MCQ 174**

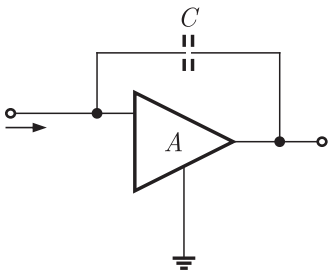
In the inverting op-amp circuit shown below, the resistance $R_g$ is chosen as $R_1 \parallel R_2$ in order to



(A) increase gain
(B) reduce offset voltage
(C) reduce offset current
(D) increase CMRR

**MCQ 175**

An amplifier of gain $A$ is bridged by a capacitance $C$ as shown below.



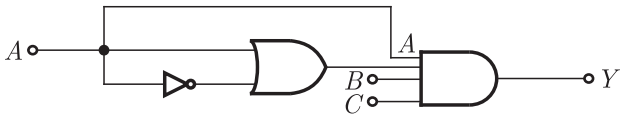

The effective input capacitance is
(A) $C/A$
(B) $C(1-A)$
(C) $C(1+A)$
(D) $CA$

**MCQ 176**

The resolution of a 12 bit Analog to Digital converter in percent is
(A) 0.01220
(B) 0.02441
(C) 0.04882
(D) 0.09760

**MCQ 177**

The Boolean expression for the output $Y$ in the logic circuit is




(A) $A\overline{B}C$
(B) $ABC$

(C) $\overline{A}BC$

(D) $\overline{A}\,\overline{B}\,\overline{C}$

**MCQ 178**

To add two m-bit numbers, the required number of half adders is

(A) $2m-1$

(B) $2^m-1$

(C) $2m+1$

(D) $2m$

**MCQ 179**

Consider the following :

Any combinational circuit can be built using

1. NAND gates
2. NOR gates
3. EX-OR gates
4. Multiplexers

Which of these are correct ?

(A) 1, 2 and 3

(B) 1, 3 and 4

(C) 2, 3 and 4

(D) 1, 2 and 4

**MCQ 180**

The decimal equivalent of hexadecimal number of 2A0F is

(A) 17670

(B) 17607

(C) 17067

(D) 10767

**MCQ 181**

The binary equivalent of hexadecimal number 4F2D is

(A) 0101 1111 0010 1100

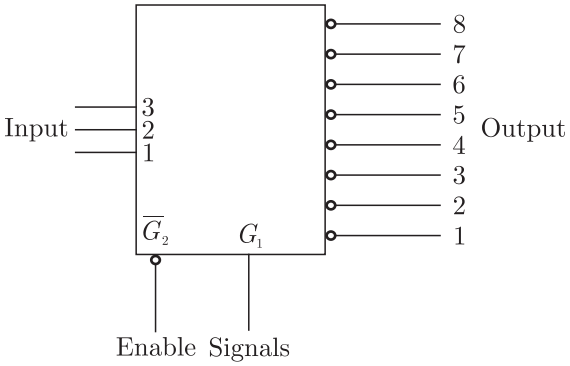
(B) 0100 1111 0010 1100

(C) 0100 1110 0010 1101

(D) 0100 1111 0010 1101

**MCQ 182**

A 3-to-8 decorder is shown below :




All the output lines of the chip will be high, when all the inputs 1, 2 and 3

(A) are high; and $G_1, G_2$ are low

(B) are high; and $G_1$ is low, $G_2$ is high

(C) are high; and $G_1, G_2$ are high

(D) are high; and $G_1$ is high, $G_2$ is low

**MCQ 183**

Which logical operation is performed by ALU of 8085 to complement a number ?

(A) AND

(B) NOT

(C) OR

(D) EXCLUSIVE OR

**Direction :** The following two items consist of two statements, one labelled as Assertion 'A' and the other labelled as Reason 'R'. You are to examine these two statements carefully and decide if the Assertion A and the Reason R are individually true and if so, whether the Reason is a correct explanation of the Assertion. Select your answers to these items using the codes given below.

**Codes :**

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

### MCQ 184

**Assertion (A) :** A tunnel diode can be used as an oscillator.

**Reason (R) :** Voltage controlled negative resistance is exhibited by a tunnel diode.

### MCQ 185

**Assertion (A) :** The intrinsic carrier concentration of Si at room temperature is more than that of GaAs.

**Reason (R) :** Si is an indirect bandgap semiconductor while GaAs is a direct bandgap semiconductor.




## YEAR 2001

**Direction :** The following four items consist of two statements, one labelled as Assertion 'A' and the other labelled as Reason 'R'. You are to examine these two statements carefully and decide if the Assertion A and the Reason R are individually true and if so, whether the Reason is a correct explanation of the Assertion. Select your answers to these items using the codes given below.

**Codes :**

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

### MCQ 186

**Assertion (A) :** In avalanche breakdown, the reverse current sharply increases with voltage due to a field emission.

**Reason (R) :** The field emission requires lightly doped $p$ and $n$

regions.

### MCQ 187

**Assertion (A) :** In small signal class 'A' amplifier, the output is a magnified replica of the input without any change in frequency.

**Reason (R) :** The dc operating point is fixed in class 'A' position.

### MCQ 188

**Assertion (A) :** D-latch and edge-trigerred D-flip flop (FF) are functionally different.

**Reason (R) :** In D-latch the output (O) cannot change while enable (EN) is High. In D-FF the output can change only on the active edge of CLK.




### MCQ 189

**Assertion (A) :** D-flip flops are used to construct a buffer register.

**Reason (R) :** Buffer registers are used to store a binary word temporarily.

### MCQ 190

A specimen of intrinsic germanium with the density of charge carriers of $2.5 \times 10^{13}/\text{cm}^3$, is doped with donor impurity atoms such that there is one donor impurity atom for every $10^6$ germanium atoms. The density of germanium atoms is $4.4 \times 10^{22}/\text{cm}^3$. The hole density would be

(A) $4.4 \times 10^{16}/\text{cm}^3$

(B) $1.4 \times 10^{10}/\text{cm}^3$

(C) $4.4 \times 10^{10}/\text{cm}^3$

(D) $1.4 \times 10^{16}/\text{cm}^3$

### MCQ 191

In a forward biased photo diode, an increase in incident light intensity causes the diode current to

(A) increase

(B) remain constant

(C) decrease

(D) remain constant while the voltage drop across the diode increases

**MCQ 192**

If for intrinsic Silicon at $27^{\circ}\mathrm{C}$, the charge concentration and mobilities of free-electrons and holes are $1.5 \times 10^{16}$per $\mathrm{m}^3$, $0.13\,\mathrm{m}^2/(\mathrm{Vs})$ and $0.05\ \mathrm{m}^2/(\mathrm{Vs})$ respectively, its conductivity will be

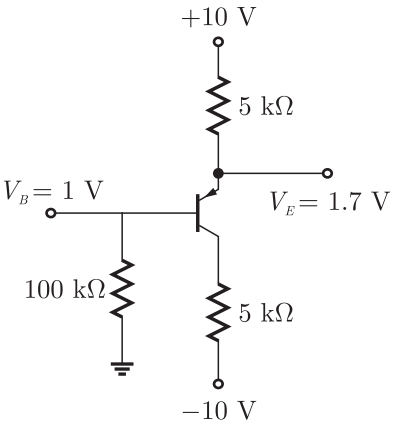
(A) $2.4 \times 10^{-3}(\Omega\mathrm{m})^{-1}$

(B) $3.15 \times 10^{-3}(\Omega\mathrm{m})^{-1}$

(C) $5 \times 10^{-4}(\Omega\mathrm{m})^{-1}$

(D) $4.32 \times 10^{-4}(\Omega\mathrm{m})^{-1}$

**MCQ 193**




A circuit using the BJT is shown in the above figure, the value of $\beta$ is

(A) 120

(B) 150

(C) 165

(D) 166

**MCQ 194**

Bridge rectifiers are preferred because

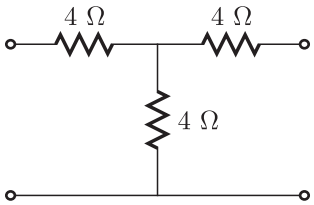
(A) they require small transformer

(B) they have less peak inverse voltage

(C) they need small transformer and also have less peak inverse voltage

(D) they have low ripple factor

**MCQ 195**




For the circuit shown in the above figure $h_{11}, h_{12}, h_{21}$ and $h_{22}$ are respectively

(A) $-0.5$, 0.5, 0.125 and 6

(B) 6, 0.5, $-0.5$, and 0.125

(C) 0.5, $-0.5$, 6 and 0.125

(D) 0.125, 6, 0.5 and $-0.5$

gate help

**MCQ 196**

In an $RC$ coupled amplifier, the gain decreases in the frequency response due to the

(A) coupling capacitor at low frequency and bypass capacitor at high frequency

(B) coupling capacitor at high frequency and bypass capacitor at low frequency

(C) coupling junction capacitance at low frequency and coupling capacitor at high frequency

(D) device junction capacitor at high frequency and coupling capacitor at low frequency

**MCQ 197**

The Darlington pair has a current gain of approximately $\beta^2$, the voltage gain $A_v$, the input resistance $R_i$ and the output resistance $R_o$. When the Darlington pair is used in the emitter-follower configuration, $A_v, R_i$ and $R_o$ are respectively

(A) very large, very large and very small

(B) unity, very large and very small

(C) unity, very small and very large

(D) very large, very small and very large

### MCQ 198

Match **List I** (Hybrid parameter) with **List II** (Units/definitions) and select the correct answer :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | $h_{ie}$ | 1. | Forward Current transfer ratio |
| B. | $h_{fe}$ | 2. | Ohms |
| C. | $h_{re}$ | 3. | Siemens |
| D. | $h_{oe}$ | 4. | Reverse Voltage transfer ratio |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 2 | 1 | 3 | 4 |
| (B) | 1 | 2 | 4 | 3 |
| (C) | 1 | 2 | 3 | 4 |
| (D) | 2 | 1 | 4 | 3 |




### MCQ 199

An amplifier having an output resistance of $4\,\Omega$ gives on open circuit output voltage of 6 V (rms). The maximum power that it can deliver to a load is

(A) 1.5 W

(B) 2.25 W

(C) 2.4 W

(D) 9 W

### MCQ 200

Active load is used in the collector of the differential amplifier of an op-amp to

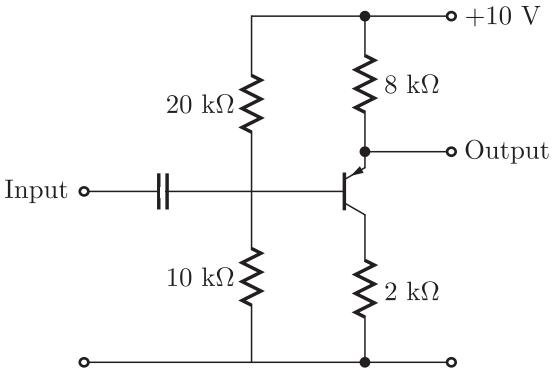
(A) increase the output resistance

(B) increase the differential gain $A_d$

(C) increase maximum peak to peak output voltage

(D) eliminate load resistance from the circuit

**MCQ 201**




For the circuit shown in the above figure, assuming $\beta = 100$ for the transistor, the transistor will be in

(A) cut off region

(B) inverse active region

(C) active region

(D) saturation region

gate help

**MCQ 202**

The slew rate of an op-amp is 0.5 V/micro sec. The maximum frequency of a sinusoidal input of 2 V rms that can be handled without excessive distortion is

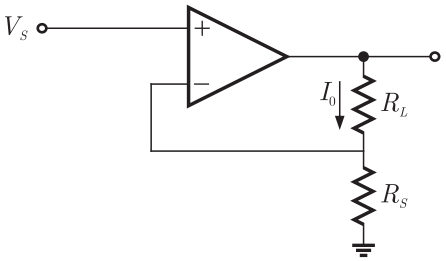
(A) 3 kHz

(B) 30 kHz

(C) 200 kHz

(D) 2 MHz

**MCQ 203**




An op-amp is used in the circuit as shown in the above figure. Current

$I_0$ is

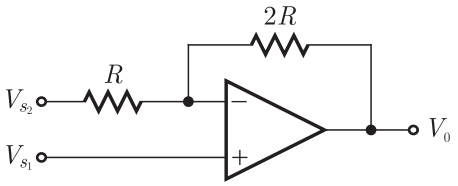
(A) $V_S \times \dfrac{R_L}{R_S(R_L + R_S)}$

(B) $V_S / R_S$

(C) $V_S / R_L$

(D) $V_S\left(\dfrac{1}{R_s} + \dfrac{1}{R_L}\right)$

**MCQ 204**




A circuit with op-amp is shown in the above figure. The voltage $V_0$ is

(A) $3\,V_{s_1} - 6\,V_{s_2}$

(B) $2\,V_{s_1} - 3\,V_{s_2}$

(C) $2\,V_{s_1} - 2\,V_{s_2}$

(D) $3\,V_{s_1} - 2\,V_{s_2}$

**MCQ 205**

A sinusoidal waveform can be converted to a square waveform by using a

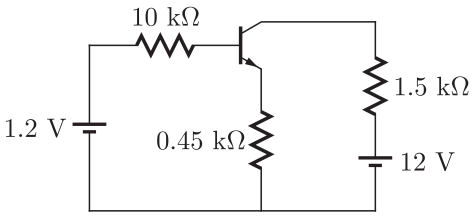
(A) two stage transistorized overdriven amplifier

(B) two stage diode detector circuit

(C) voltage comparator based on op-amp

(D) regenerative voltage comparator circuit

**MCQ 206**




For the circuit shown in the above figure, by assuming $\beta = 200$ and

$V_{BE} = 0.7\ V$, the best approximation for the collector current $I_c$ in the active region is

(A) 1 mA

(B) 2.4 mA

(C) 3 mA

(D) 9.6 mA

**MCQ 207**

High power efficiency of the push-pull amplifier is due to the fact that

(A) each transistor conducts on different cycle of the input

(B) transistors are placed in CE configuration

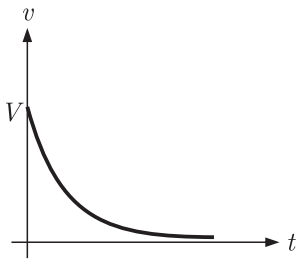
(C) there is no quiescent collector current

(D) low forward biasing voltage is required

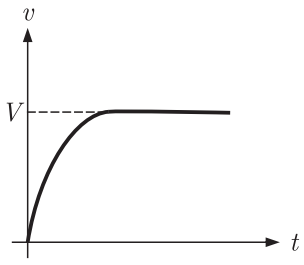
**MCQ 208**

Which one of the following is the output of the high pass filter to a step input ?

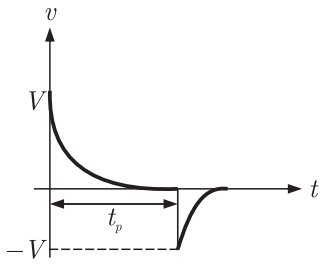
(A)



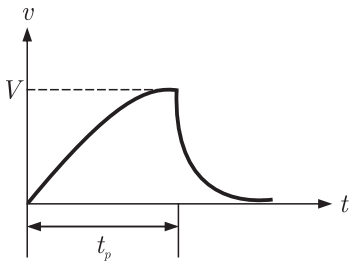

(B)



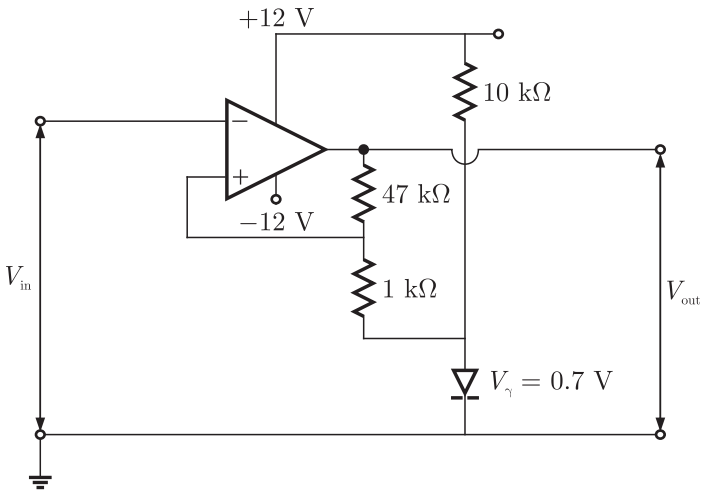

(C)




(D)




**MCQ 209**




The Schmitt trigger circuit is shown in the above figure. If $V_{sat} = \pm$ 10 V , the tripping point for the increasing input voltage will be

(A) 1 V

(B) 0.893 V

(C) 0.477 V

(D) 0.416 V

gate help

**MCQ 210**

In Boolean Algebra,

If $F = (A + B)(\overline{A} + C)$, then

(A) $F = AB + \overline{A}C$

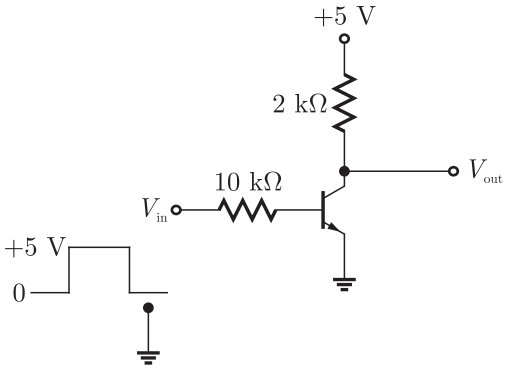
(B) $F = AB + \overline{AB}$

(C) $F = AC + \overline{A}B$

(D) $F = AA + \overline{A}B$

**MCQ 211**

A switch circuit using the transistor is shown in the figure below. Assume $h_{FE(min)} = 20$ and $f_T = 100$ MHz. The most dominant speed limitation is brought by




(A) rise time

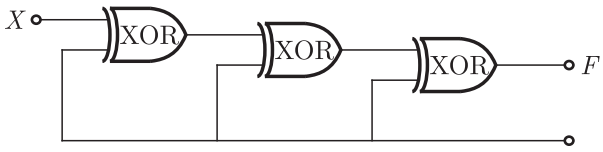
(B) fall time

(C) storage time

(D) delay time

gate help

**MCQ 212**




For the circuit shown in the above figure, the output $F$ will be

(A) 1

(B) Zero

(C) $X$

(D) $\overline{X}$

**MCQ 213**

In the CMOS inverter, the power dissipation is

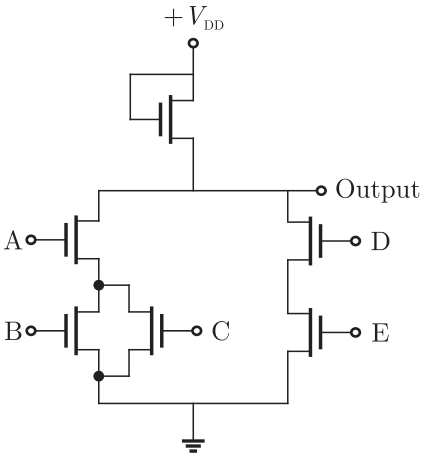
(A) low only when $V_{IN}$ is low

(B) low only when $V_{IN}$ is high

(C) high during dynamic operation

(D) low during dynamic operation

**MCQ 214**




An NMOS circuit is shown in the above figure. The logic function for the output (o/p) is

(A) $(\overline{A+B}).C+\overline{D}.\overline{E}$

(B) $(AB+\overline{C}).(\overline{D}+E)$

(C) $\overline{A.(B+C)+D.E}$

(D) $\overline{ABCDE}$

gate help

**MCQ 215**

Match **List I** (Types of gates) with **List II** (Values of propagation delay) and select the correct answer :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | ECL | 1. | 5 ns |
| B. | TTL | 2. | 20 ns |
| C. | CMOS | 3. | 100 ns |
| D. | NMOS | 4. | 1 ns |

**Codes :**

| | A | B | C | D |
|---|---|---|---|---|
| (A) | 1 | 4 | 3 | 2 |
| (B) | 4 | 1 | 3 | 2 |
| (C) | 1 | 4 | 2 | 3 |
| (D) | 4 | 1 | 2 | 3 |